

Agent Acceleration in a Stable Environment

Learning from repeated tasks to make later ones faster and cheaper

1. Can what an agent learns on earlier tasks lower the time and money of later tasks, with the cost of learning counted?
2. After how many tasks does the learning pay for itself?
3. Which parts of the agent should learning change, and should it learn from its own runs or from exploring the environment?
4. Does what it learned still work on new tasks and after the environment changes, without lowering the success rate?

Problem: make repeated tasks faster and cheaper

An agent runs the same workflow many times with new inputs. What it learns on earlier tasks should lower the time and money of later tasks, with the cost of learning counted, without lowering the success rate.

FORMULA	MEANING	SYMBOLS	SOURCE
$v(m, p) = \frac{C_m(p)}{R_m(p)}$ $T_{\text{success}}(m, p) = \frac{\mathbb{E}[T_{\text{attempt}}]}{R_m(p)}$	v (the source’s cost-of-pass): expected cost per success on one task, retrying until it succeeds T_{success} : expected time per success on one task	m : the agent, a model plus the program around it (the harness); “m” for model, as in the source p : one task, e.g. one expense report; “p” for problem, as in the source $C_m(p)$: expected cost (C) of one attempt, in dollars $R_m(p)$: success rate (R), the chance that one attempt succeeds T_{attempt} : time (T) of one attempt, in seconds	v : from Erol et al. (2026) T_{success} : self-defined, the same form as v
$\bar{v}(m) = \frac{1}{n} \sum_{k=1}^n v(m, p_k)$	$\bar{v}$: the average over n tasks; $\bar{T}_{\text{success}}$ and the average success rate $\bar{R}$ are taken the same way	n : number of tasks, e.g. expense reports filed p_k : the k -th task, $k = 1, \dots, n$	Self-defined, adapted from Erol et al. (2026).
$v_n(m') = \frac{C_{\text{learn}}}{n} + \bar{v}(m')$	v_n : expected cost per task after learning, learning cost included	m' : the same agent after learning, model unchanged, harness changed C_{learn} : cost of learning (exploring, proposing and checking changes, keeping them up to date), each counted once	Self-defined, adapted from Kapoor et al. (2025) and Cai et al. (2024).
$\min (\bar{T}_{\text{success}}(m'), v_n(m'))$ s. t. $\bar{R}(m') \geq R_0$	The goal: lower time and money per task, success held	R_0 : lowest required success rate, fixed in advance, measured on tasks not used for learning	Self-defined.
$n^* = \frac{C_{\text{learn}}}{\bar{v}(m) - \bar{v}(m')}$	n^* : break-even, the number of tasks after which $v_n(m')$ falls below $\bar{v}(m)$	—	Self-defined, from row 3; Kapoor et al. (2025) report a break-even point.

Each source’s original formula, and what we changed and why: [Appendix A0 ↗](#)

REFERENCES

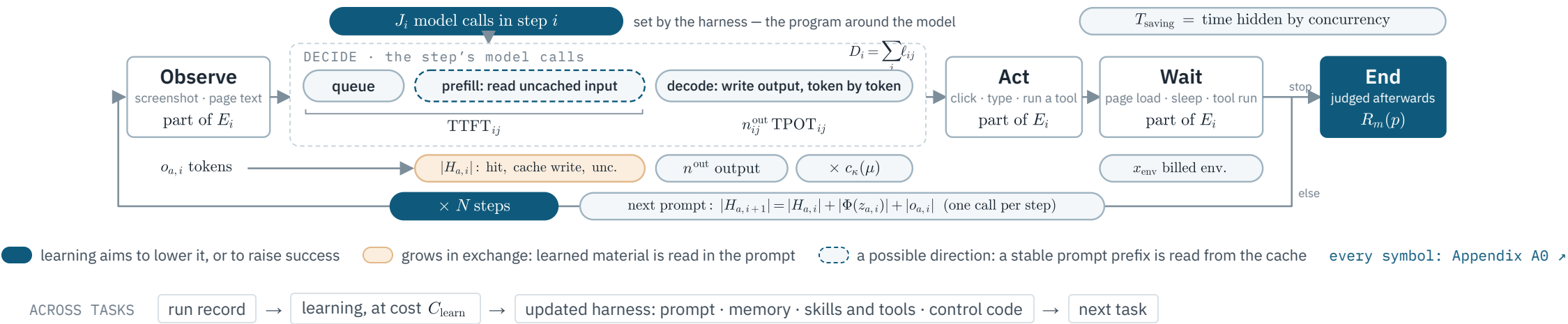
• Cai, T., Wang, X., Ma, T., et al. (2024). Large Language Models as Tool Makers. *International Conference on Learning Representations 2024 (ICLR 2024)*.

• Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.

• Kapoor, S., Stroebl, B., Siegel, Z. S., et al. (2025). AI Agents That Matter. *Transactions on Machine Learning Research*.

How an agent runs a task, and where learning changes it

One attempt is a loop of steps: observe, call the model, act, wait, until the agent stops. Learning across tasks changes the harness around the model to cut steps, model calls and failures; the price is a longer prompt to read.



- N , i steps in one attempt; one step is one observe–decide–act–wait round (Hua et al., 2025)
- J_i model calls in step i : planner, judge, retries (self-defined)
- $R_m(p)$ chance that one attempt of agent m on task p succeeds (Erol et al., 2026)
- $|H_{a,i}|$ length of the prompt read in step i ; learned notes make it longer (Y. Yuan et al., 2026)
- prefill reading the prompt before writing; input already in the cache is read faster and billed at a lower price (Anthropic, 2026b)

REFERENCES

Anthropic. (2026b). Pricing; Prompt caching; Fast mode (research preview); Vision; Steering thinking; Migrating to Claude Opus 5.5 (developer documentation, read 28–29 September 2026). <https://platform.claude.com/docs/en/about-claude/pricing>; <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>; <https://platform.claude.com/docs/en/build-with-claude/fast-mode>; <https://platform.claude.com/docs/en/build-with-claude/vision>; <https://platform.claude.com/docs/en/build-with-claude/thinking-steering-and-cost>; <https://platform.claude.com/docs/en/models/opus-5-5/migration-guide>

Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.

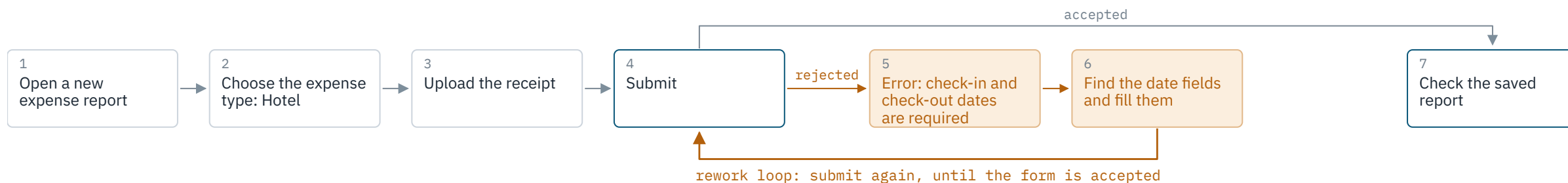
Hua, W., Wan, M., Vadrevu, S., Nadel, R., Zhang, Y., & Wang, C. (2025). Interactive speculative planning: Enhance agent efficiency through co-design of system and user interface (ISP). *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.00079. Rutgers University; Microsoft; Google DeepMind.

Yuan, Y., Nayak, A., Kundu, S., & Talati, N. (2026). Agentic AI workload characterization. *Proceedings of the 2026 IEEE International Symposium on Workload Characterization (IISWC 2026)*, Boulder, CO. arXiv:2605.26297v2 (21 September 2026; arXiv title “Agentic AI workload characteristics”). University of Illinois Urbana-Champaign; Gimlet Labs; Intel.

Example: a web agent that files expense reports

Our running example: a web agent files expense reports; each task is a new report in the same system. Not knowing one rule of the form, that a hotel expense needs check-in and check-out dates, sends the agent into a rework loop.

ONE HOTEL REPORT, BEFORE THE AGENT KNOWS THE RULE



WITHOUT LEARNING

The loop is paid again on every hotel report

- Its extra model calls and seconds are part of $\bar{v}(m)$, for each of the n reports

LEARNING THE RULE ONCE

The rule is learned once, at cost C_{learn} ; later reports skip the loop

- $\bar{v}(m') < \bar{v}(m)$; it pays off after n^* reports (page 1)

IN THIS EXAMPLE

- **Environment:** the expense system, its forms, the fields each expense type requires, the actions it allows
- **Run record:** the steps above, with the error message, and the time and cost of each step

WHAT LEARNING MAY CHANGE HERE (MODEL WEIGHTS FIXED)

- **Prompt:** an instruction read on every call: fill the check-in and check-out dates before submitting a hotel expense
- **Memory:** a note read only when relevant: Hotel → check-in and check-out dates required
- **Skill or tool:** a procedure the model calls when it needs it: instructions plus a script that fills every field a hotel expense requires
- **Control code:** the harness's own logic, run whatever the model decides: check the required fields before every submit

SCOPE

this example: expense reports on a website



any workflow repeated in one software environment



websites · desktop applications · mobile apps · tool APIs

Five difficulties, grouped by the cost they affect (version A)

Three difficulties affect the cost of each task after learning; two affect the cost of learning itself.

COST PER TASK AFTER LEARNING · $\bar{v}(m')$			
1 · Fewer steps do not always save time	WebCoach (Liu et al., 2025) actions 10.7 → 10.2, but 215 → 395 s per task	ReasoningBank (Ouyang et al., 2026) steps 9.7 → 8.3, but total tokens 50,847 → 53,055 in another table	GenericAgent (Liang et al., 2026) fewest tokens on some tasks, yet slower than OpenClaw on five long tasks: 220.8 vs 183.1 s
2 · What can be skipped is known only after a run has succeeded	EchoPath (Zhao et al., 2026) 315.7 → 127.5 s per task, only for 159 tasks already solved; storing one took ~572k tokens	MobileGPT (Lee et al., 2024) −62.5% latency, −68.8% cost on repeated tasks, after 10–15 min of exploring each app	ActionEngine (Zhong et al., 2026) 91.2% with a warmed-up site map, 73.1% without; exploring pays off after 39–101 tasks
3 · What was learned must help on new tasks and be cheap to read	Skill and memory modules (Hajimiri et al., 2026) with budgets roughly matched, the plain agent did best: 44.78% at 73.6K tokens per task	ClawTrace (Yuan et al., 2026) moved to SkillsBench, median cost per task \$0.143 vs \$0.144: no saving	SEDM (Xu et al., 2025) its memory raised the prompt on FEVER to 2.47M tokens, from 1.65M without memory
COST OF LEARNING · C_{learn}			
4 · Finding the right change from a failure is costly, and not guaranteed	HarnessFix (Chen et al., 2026) completion +6.3 to +18.4 points, for 37.2M tokens of repairs on AppWorld	ESPO (L. Liu et al., 2026) about 3.4M tokens and 2 h 13 min to optimise one prompt (PUPA)	GEPA (Agrawal et al., 2026) 1,839–7,051 rollouts to optimise the prompts of one benchmark
5 · Learning, with its testing, can cost more than the mistakes it removes	SICA (Robeyns et al., 2025) about \$7,000 for one full run, to save \$0.21 per task (calc.)	ADAS (Hu et al., 2025) about \$500 for one search and its evaluation on ARC	GEA (Weng et al., 2026) about \$13,000 for one full evolution (author estimate)

REFERENCES

- Agrawal, L. A., Tan, S., Soylu, D., et al. (2026). GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. *ICLR 2026*.
- Chen, M., Wang, J., Liu, Z., et al. (2026). From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint*.
- Hajimiri, S., Aminbeidokhti, M., Dolz, J., et al. (2026). Are Online Skill and Memory Modules Always Worth Their Tokens? A Budget-Constrained Study of Web Agents. *EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing)*.
- Hu, S., Lu, C., & Clune., J. (2025). Automated Design of Agentic Systems. *International Conference on Learning Representations 2025 (ICLR 2025)*.
- Lee, S., Choi, J., Lee, J., et al. (2024). MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom 2024)*.

- Liang, J., Han, J., Li, W., et al. (2026). GenericAgent: A Token-Efficient Self-Evolving LLM Agent via Contextual Information Density Maximization (V1.0). *arXiv preprint*.
- Liu, G., Geng, S., Li, S., et al. (2025). WebCoach: Self-Evolving Web Agents with Cross-Session Memory Guidance. *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents)*.
- Liu, L., Tang, P., Singh, K. Y., et al. (2026). ESPO: Error-Structured Prompt Optimization via Diagnose, Diversify, and Stabilize. *EMNLP 2026*.
- Ouyang, S., Yan, J., Hsu, I. H., et al. (2026). ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. *ICLR 2026*.
- Robeyns, M., Szummer, M., & Aitichison., L. (2025). A Self-Improving Coding Agent. *ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision'*.
- Weng, Z., Antoniadis, A., Nathani, D., et al. (2026). Group-Evolving Agents: Open-Ended Self-Improvement via Experience Sharing. *arXiv preprint*.

- Xu, H., Hu, J., Zhang, K., et al. (2025). SEDM: Scalable Self-Evolving Distributed Memory for Agents. *SEA @ NeurIPS 2025 (NeurIPS 2025 workshop)*.
- Yuan, B., Su, Y., Song, R., et al. (2026). ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation. *Agent Skills '26 Workshop at ACM CAIS 2026 (ACM Conference on AI and Agentic Systems), San Jose, May 26, 2026*.
- Zhao, Y., Shanmugham, A., Roy, S., et al. (2026). EchoPath: Execution-Level Replayable Memory for GUI Agents. *arXiv preprint*.
- Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*.

Five difficulties, explained (version B)

COST PER TASK AFTER LEARNING · $\bar{v}(m')$

1 · Fewer steps do not always save time

- A learned hint, or a second model that coaches, adds reading and thinking to every step
- Steps differ in cost: skipping two clicks saves little if each remaining step waits longer
- So judge a report by its seconds and dollars, not by its step count

Works: WebCoach (Liu et al., 2025) · ReasoningBank (Ouyang et al., 2026) · GenericAgent (Liang et al., 2026)

2 · What can be skipped is known only after a run has succeeded

- Before the first hotel report goes through, the agent cannot tell the rework from the necessary steps
- Savings start once a successful run is stored, and storing it costs a full first pass
- A kind of report never seen before still starts from scratch

Works: EchoPath (Zhao et al., 2026) · MobileGPT (Lee et al., 2024) · ActionEngine (Zhong et al., 2026)

3 · What was learned must help on new tasks and be cheap to read

- A note learned on hotel reports may not help with meal or travel reports
- Every note in the prompt is read again on every call; too many notes cost more than they save
- A misleading note can even lower the success rate

Works: skill and memory modules (Hajimiri et al., 2026) · ClawTrace (Yuan et al., 2026) · SEDM (Xu et al., 2025)

COST OF LEARNING · C_{learn}

4 · Finding the right change from a failure is costly, and not guaranteed

- The run record shows that a report was rejected, not whether the prompt, a skill or a check should change
- Diagnosing, writing a change and re-running tasks all cost model calls
- The change found can be wrong, fix only some failures, or break other tasks

Works: HarnessFix (Chen et al., 2026) · ESPO (L. Liu et al., 2026) · DRAFT (Qu et al., 2025)

5 · Learning, with its testing, can cost more than the mistakes it removes

- To trust a change, it must be re-run on many reports, and those runs cost money
- Without learning, the agent simply pays for its rework on each report
- Learning pays off only after enough reports: the break-even n^* of page 1

Works: SICA (Robeyns et al., 2025) · ADAS (Hu et al., 2025) · GEA (Weng et al., 2026)

REFERENCES

- Chen, M., Wang, J., Liu, Z., et al. (2026). From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint*.
- Hajimiri, S., Aminbeidokhti, M., Dolz, J., et al. (2026). Are Online Skill and Memory Modules Always Worth Their Tokens? A Budget-Constrained Study of Web Agents. *EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing)*.
- Hu, S., Lu, C., & Clune, J. (2025). Automated Design of Agentic Systems. *International Conference on Learning Representations 2025 (ICLR 2025)*.
- Lee, S., Choi, J., Lee, J., et al. (2024). MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom 2024)*.
- Liang, J., Han, J., Li, W., et al. (2026). GenericAgent: A Token-Efficient Self-Evolving LLM Agent via Contextual Information Density Maximization (V1.0). *arXiv preprint*.

- Liu, G., Geng, S., Li, S., et al. (2025). WebCoach: Self-Evolving Web Agents with Cross-Session Memory Guidance. *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents)*.
- Liu, L., Tang, P., Singh, K. Y., et al. (2026). ESPO: Error-Structured Prompt Optimization via Diagnose, Diversify, and Stabilize. *EMNLP 2026*.
- Ouyang, S., Yan, J., Hsu, I. H., et al. (2026). ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. *ICLR 2026*.
- Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025)*.
- Robeyns, M., Szummer, M., & Aitchison, L. (2025). A Self-Improving Coding Agent. *ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision'*.
- Weng, Z., Antoniadis, A., Nathani, D., et al. (2026). Group-Evolving Agents: Open-Ended Self-Improvement via Experience Sharing. *arXiv preprint*.

- Xu, H., Hu, J., Zhang, K., et al. (2025). SEDM: Scalable Self-Evolving Distributed Memory for Agents. *SEA @ NeurIPS 2025 (NeurIPS 2025 workshop)*.
- Yuan, B., Su, Y., Song, R., et al. (2026). ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation. *Agent Skills '26 Workshop at ACM CAIS 2026 (ACM Conference on AI and Agentic Systems), San Jose, May 26, 2026*.
- Zhao, Y., Shanmugham, A., Roy, S., et al. (2026). EchoPath: Execution-Level Replayable Memory for GUI Agents. *arXiv preprint*.
- Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*.

Five difficulties (version C: diagrams)

COST PER TASK AFTER LEARNING · $\bar{v}(m')$

1 · Fewer steps do not always save time

BEFORE

AFTER

one step

extra reading and thinking in each step

length = seconds per task · per step 20.1 → 38.7 s (calc.)

Each step got longer: judge a report by its seconds, not its steps

Drawn from WebCoach (Liu et al., 2025). Also: ReasoningBank (Ouyang et al., 2026), fewer steps, more tokens

2 · Whether a check is needed is not known in advance

Choose type: Hotel

check the form

fill dates

submit

one step more, always right

skip: as last time

submit

rule still holds

it changed: rework

Skipping a check saves a step only while the rule still holds

Seen in: DRAFT (Qu et al., 2025), learns tool conditions by trying · SKILL.nb (Hattami et al., 2026), checks before reuse

3 · A trade-off between generalization and cost per use

generalizes ↑

cheap and general?

note in every prompt

note in memory

skill: steps + script

replayed script

schematic

cost per use →

Seen in: Metis (Dai et al., 2026), keeps both text and code · ActionEngine (Zhong et al., 2026), turns a site map into programs

COST OF LEARNING · C_{learn}

4 · Finding the cause behind an error message

Error: over the meal limit

possible causes, of different kinds, checked one by one

reasoning wrong type?

rule limit changed?

interaction wrong field?

tool amount misread?

...

the cause may be found late, or not at all

cost of checking so far →

Each possible cause costs a check; the right one may come late, or never

Seen in: HarnessFix (Chen et al., 2026), finds the faulty step · ESPO (L. Liu et al., 2026), sorts errors first

5 · Testing a change costs money, and can cost more than it saves

total cost ↑

learning and testing are paid first

break-even n^*

break-even, more testing

repair after the site changes

no learning more testing with learning

schematic

number of reports filed →

Learning pays off only if enough reports share its cost

Seen in: AI Agents That Matter (Kapoor et al., 2025), late break-even · SICA (Robeyns et al., 2025), large bill, small saving

REFERENCES

• Chen, M., Wang, J., Liu, Z., et al. (2026). From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint*.

• Dai, Z., He, S., Li, H., et al. (2026). Metis: Bridging Text and Code Memory for Self-Evolving Agents. *arXiv preprint*.

• Hattami, A. E., Chapados, N., & Pal, C. (2026). SKILL.nb: Selective Formalization and Gated Execution for Durable Agent Workflows. *FAGEN@ICML 2026 Poster (ICML 2026 workshop, venueid ICML.cc/2026/Workshop/FAGEN)*.

• Kapoor, S., Stroebel, B., Siegel, Z. S., et al. (2025). AI Agents That Matter. *Transactions on Machine Learning Research*.

• Liu, G., Geng, S., Li, S., et al. (2025). WebCoach: Self-Evolving Web Agents with Cross-Session Memory Guidance. *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents)*.

• Liu, L., Tang, P., Singh, K. Y., et al. (2026). ESPO: Error-Structured Prompt Optimization via Diagnose, Diversify, and Stabilize. *EMNLP 2026*.

• Ouyang, S., Yan, J., Hsu, I. H., et al. (2026). ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. *ICLR 2026*.

• Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025)*.

• Robeyns, M., Szummer, M., & Aitchison, L. (2025). A Self-Improving Coding Agent. *ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision'*.

• Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*.

Direction 1 · Learning from the agent’s own runs: 93 works in four classes

Of 93 works that change what sits outside the model, 21 measured the seconds or dollars of running tasks, 19 what learning cost, and 28 tested on unseen tasks.

PER TASK $v_n(m') = \frac{C_{\text{learn}}}{n} \uparrow + \bar{v}(m') \downarrow$ s. t. $\bar{R}(m') \geq R_0$

CLASS	WORKS	WHAT CHANGES	REPRESENTATIVE WORK: ONE NUMBER AND ITS CONDITIONS	TIME OR MONEY OF RUNS YES · PARTLY	LEARNING COST YES · PARTLY	UNSEEN TASKS YES · PARTLY
Prompts and context A02 ↗	15	the prompt, instructions or context read on the next task	ACE: offline adaptation lifts AppWorld from 42.4 to 59.4 (with labels), and takes 9,517 s against GEPA’s 53,898 s — DeepSeek-V3.1 (Q. Zhang et al., 2026)	1 · 2	5 · 5	7 · 6
Experience memory A03 ↗	26	retrievable experience: successes, failures, workflows	AWM: 35.5% vs 23.5% success on WebArena (the baseline also reads HTML), 5.9 vs 7.9 steps against an accessibility-tree BrowserGym — GPT-4-0613 (Wang et al., 2025a)	5 · 15	5 · 4	11 · 14
Skills and tools A04 ↗	34	callable functions, scripts, skill documents	AXIS: 29.9 s vs 59.5 s and \$0.2 vs \$0.4 per task, 84% vs 52% success, against UFO on 50 Microsoft Word tasks; learning cost not reported (Lu et al., 2025)	9 · 14	6 · 11	8 · 23
Whole harness A05 ↗	30	the outer program: tools, control loop, the improver itself	SICA: \$1.91 → \$1.70 and 130.2 → 114.5 s per task, for a whole run of about \$7,000; the benchmark that picked the agent also scored it (Robeyns et al., 2025)	8 · 4	7 · 16	5 · 23
All, each work once	93			21 · 28	19 · 33	28 · 57

Yes · partly: our initial judgment from the per-work table (calc.; Appendix A1). A work in two classes counts in both.

REFERENCES

- Lu, J., Zhang, Z., Yang, F., et al. (2025). AXIS: Efficient human-agent-computer interaction with API-first LLM-based agents. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Robeyns, M., Szummer, M., & Aitchison., L. (2025). A Self-Improving Coding Agent. *ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision'*.

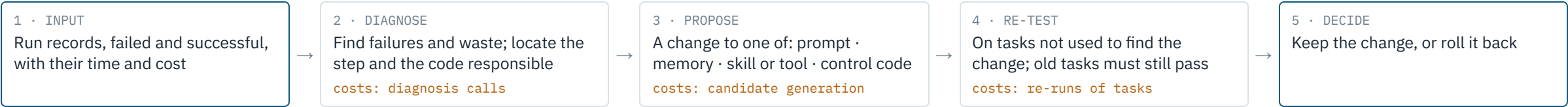
- Wang, Z. Z., Mao, J., Fried, D., et al. (2025). Agent Workflow Memory. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, PMLR 267:63897-63911, 2025 (the PMLR page names no track).
- Zhang, Q., Hu, C., Upasani, S., et al. (2026). Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models. *ICLR 2026*.

How a run record becomes a change, and what the change costs

Every step after reading the run records costs model calls, and these costs are rarely reported next to the savings.

PER TASK $v_n(m') = \frac{C_{\text{learn}}}{n} \uparrow + \bar{v}(m') \downarrow \quad \text{s. t.} \quad \bar{R}(m') \geq R_0$

targeted in exchange unchanged



IT WORKS

Changes can be found and made automatically, with gains on the final tasks

- Highest success in five of six settings and 76.0–91.8% fewer model calls at deployment — Growing Harness, BrowseComp-Plus and WebArena-Verified, three models, 50 final tasks per setting, three runs (L. Li et al., 2026)

LEARNING COST LEFT OUT · C_{learn}

Savings at deployment are usually reported without the learning that produced them

- 97.4K vs 112.6K tokens and 11.25 vs 14.55 turns per AppWorld task once the memory is frozen; the memory manager’s own calls are not included — Metis (Dai et al., 2026)

REFERENCES

- Dai, Z., He, S., Li, H., et al. (2026). Metis: Bridging Text and Code Memory for Self-Evolving Agents. *arXiv preprint*.
- Li, L., Li, J., Zhao, J., et al. (2026). Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents. *arXiv preprint*.

Direction 2 · Learning the environment: 29 works keep four kinds of knowledge

Of 29 works that keep knowledge about the environment, 14 measured the seconds or dollars of running tasks, 5 what exploring cost, and 8 tested on unseen tasks.

PER TASK $v_n(m') = \frac{C_{\text{learn}}}{n} \uparrow + \bar{v}(m') \downarrow$ s. t. $\bar{R}(m') \geq R_0$

CLASS	WORKS	WHAT CHANGES	REPRESENTATIVE WORK: ONE NUMBER AND ITS CONDITIONS	TIME OR MONEY OF RUNS YES · PARTLY	LEARNING COST YES · PARTLY	UNSEEN TASKS YES · PARTLY
Predicting what an action does A06 ↗	3	what the page will look like after an action	WMA: 140.3 s vs 748.3 s and \$0.4 vs \$2.7 per instruction against tree search, 16.6% vs 19.2% success — WebArena, GPT-4o (Chae et al., 2025)	1 · 1	1 · 2	1 · 2
Descriptions, facts and preconditions A07 ↗	13	tool parameters and errors; facts and traps of the site	DRAFT: correct tool paths 88.00 vs 71.00 for ReAct on RestBench TMDB, GPT-4o; time and cost not reported (Qu et al., 2025)	6 · 4	2 · 3	3 · 8
Page and place structure A08 ↗	7	a map of pages, actions and the paths between them	MobileGPT: −62.5% latency and −68.8% cost on repeated tasks with human-repaired paths; exploring took 10–15 min per app, \$10.78 in all (Lee et al., 2024)	5 · 2	1 · 3	1 · 6
Exploring and practising A09 ↗	9	self-set practice tasks, turned into tools	WALT: 50.1% and 52.9% success on WebArena and VisualWebArena; exploring and checking cost \$1.67 per tool, repaid after about 14 uses (Prabhu et al., 2026)	3 · 2	1 · 3	4 · 4
All, each work once	29			14 · 8	5 · 10	8 · 18

Yes · partly: our initial judgment from the per-work table (calc.; Appendix A1). A work in two classes counts in both.

REFERENCES

- Chae, H., Kim, N., Ong, K., et al. (2025). Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation. *International Conference on Learning Representations 2025 (ICLR 2025)*.
- Lee, S., Choi, J., Lee, J., et al. (2024). MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom 2024)*.

- Prabhu, V., Dai, Y., Fernandez, M., et al. (2026). WALT: Web Agents that Learn Tools. *International Conference on Learning Representations 2026 (ICLR 2026)*.
- Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025)*.

Exploring a site, keeping a checked rule, and using it on the next task

The agent probes the form, writes down what it found together with where it applies and the evidence for it, and reads that note on the next task, so the rework loop of the expense example does not happen.

1 · EXPLORE THE FORM

Change the expense type from Meal to Hotel. Two new required fields appear: check-in and check-out date. Submitting without them returns an error.

2 · KEEP A CHECKED RULE

site-guide.md

scope: new expense form
IF type = Hotel → check-in and check-out dates required
evidence: 2 probes, 1 error message
re-check: when the form changes

3 · USE IT ON THE NEXT TASK

New hotel expense + the live page + the saved note → fill every required field → submit once → check the saved report.



REAL CALLS CORRECT THE DESCRIPTION

Trying an action and reading its result can fix what a description gets wrong

- DRAFT learned from a real call that a tool needs a valid person_id, and wrote it into the tool’s documentation (Qu et al., 2025)

ONE OBSERVATION IS NOT A RULE

A rule written from one observation can be wrong, and a local check can miss it

- DRAFT generalised from one error message; SkillWeaver’s local checks passed broken functions (Qu et al., 2025; Zheng et al., 2025)

REFERENCES

- Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025)*.
- Zheng, B., Fatemi, M. Y., Jin, X., et al. (2025). SkillWeaver: Web Agents can Self-Improve by Discovering and Honing Skills. *arXiv preprint*.

Across both directions, what learning costs is rarely counted

The learning bill is rarely itemised, proving that a change is correct is the weak link, and a saving measured as fewer steps, or in one setting, does not guarantee less time or money elsewhere.

LEARNING COST · C_{learn}

Few works count both what learning costs and what it saves

- 5 of 102 works measured both the seconds or dollars of running tasks and a learning cost: GPTSwarm, MobileGPT, OpenSkill, EchoPath and Unbrowse (initial judgment, calc.) (Zhuge et al., 2024; Lee et al., 2024; Yan et al., 2026; Zhao et al., 2026; Tham et al., 2026)

CORRECTNESS · $\bar{R}(m') \geq R_0$

Showing that a change fixed the target without breaking what worked is the open problem

- In AgentDevel’s ablations, the highest mean score and the fewest regressions on old tasks may have to be traded against each other (Zhang, 2026)

FEWER STEPS · $\bar{v}(m')$

Fewer steps is not the same as less time or money

- Steps 9.7 → 8.3 on WebArena, while total tokens rose 50,847.4 → 53,054.5 in another table whose base set-up is not stated — ReasoningBank (Ouyang et al., 2026)

TRANSFER · $\bar{v}(m')$

A saving in one setting need not carry over

- Median cost per task \$0.143 vs \$0.144 after moving to SkillsBench, no fall — ClawTrace, OpenClaw with gpt-5.4 (Yuan et al., 2026)

REFERENCES

- Lee, S., Choi, J., Lee, J., et al. (2024). MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom 2024)*.
- Ouyang, S., Yan, J., Hsu, I. H., et al. (2026). ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. *ICLR 2026*.
- Tham, L., Garcia, N. M. G., & Hahn, J. (2026). Internal APIs Are All You Need: Shadow APIs, Shared Discovery, and the Case Against Browser-First Agent Architectures. *arXiv preprint*.
- Yan, Z., Song, D., Zhang, H., et al. (2026). OpenSkill: Open-World Self-Evolution for LLM Agents. *EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing)*.

- Yuan, B., Su, Y., Song, R., et al. (2026). ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation. *Agent Skills '26 Workshop at ACM CAIS 2026 (ACM Conference on AI and Agent Systems)*, San Jose, May 26, 2026.
- Zhang, D. (2026). AgentDevel: Reframing Self-Evolving LLM Agents as Release Engineering. *arXiv preprint*.
- Zhao, Y., Shanmugham, A., Roy, S., et al. (2026). EchoPath: Execution-Level Replayable Memory for GUI Agents. *arXiv preprint*.
- Zhuge, M., Wang, W., Kirsch, L., et al. (2024). GPTSwarm: Language Agents as Optimizable Graphs. *ICML 2024: Proceedings of the 41st International Conference on Machine Learning, PMLR 235:62743-62767 (2024)*.

The closest works save time or money on repeated tasks; none counts the whole loop

In the works we checked, no single result covers all of these at once: any task stream, no training, automatic location of failures and waste, protection of old abilities, every learning and maintenance cost counted, and a net saving over time.

WORK	WHAT IT LEARNS	WHAT IT SHOWS: ONE NUMBER AND ITS CONDITIONS	WHAT IS LEFT OUT
ActionEngine (Zhong et al., 2026)	a state-machine map of the site, with templates	91.2% success (73.1% without the warmed-up map); 27 s vs 87 s and \$0.05 vs \$0.40 per task against Claude Code — WebArena, 655 tasks, Claude Opus 4.6	per-task numbers exclude exploration and warm-up; break-even after 39–101 tasks still omits warm-up
WALT (Prabhu et al., 2026)	tools written from exploring each site	\$1.67 per tool to explore and check it, repaid after about 14 uses	repairing tools after deployment is left to future work
SpeedRunner (Huang et al., 2026)	programmatic skills	BabyAI usage about one eighth, success from about 67% to near-perfect — gpt-5.4-mini, 200 online tasks, 30 held out, three seeds	output tokens and dollars disagree in one figure; total learning cost [TBD: appendix A.3, to be re-read]
HarnessFix (Chen et al., 2026)	repairs to the outer program from failed runs	completion +6.3 to +18.4 points, mean of three runs	time and cost of running tasks not reported
Growing Harness (L. Li et al., 2026)	code changes from function-level failures	inference cost at deployment −74.4% to −98.6%	evaluator and offline optimiser not counted; break-even not measured
Metis (Dai et al., 2026)	text and code memory, frozen before the final tasks	97.4K vs 112.6K tokens per AppWorld task	memory manager not counted; apps never seen before not tested

REFERENCES

- Chen, M., Wang, J., Liu, Z., et al. (2026). From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint*.
- Dai, Z., He, S., Li, H., et al. (2026). Metis: Bridging Text and Code Memory for Self-Evolving Agents. *arXiv preprint*.
- Huang, Z., Wang, X., Wang, A., et al. (2026). Better, Faster, Stronger: Programmatic Skill Learning Best Reduces Agent Cost. *arXiv preprint*.
- Li, L., Li, J., Zhao, J., et al. (2026). Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents. *arXiv preprint*.

- Prabhu, V., Dai, Y., Fernandez, M., et al. (2026). WALT: Web Agents that Learn Tools. *International Conference on Learning Representations 2026 (ICLR 2026)*.
- Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*.

References (1/6)

Agashe, S., Han, J., Gan, S., et al. (2025). Agent S: An Open Agentic Framework that Uses Computers Like a Human. *International Conference on Learning Representations 2025 (ICLR 2025), main conference, poster (decision 'Accept (Poster)'*. Formal: [link](#). Version read: [HTML](#), v1, 2024-10-10.

Agrawal, L. A., Tan, S., Soylu, D., et al. (2026). GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. *ICLR 2026, main conference*. Formal: [link](#). Version read: [HTML](#), v2, 2026-02-14.

Amayuelas, A., Laakom, F., Piękos, P., et al. (2026). Planning to Explore: Curiosity-Driven Planning for LLM Test Generation. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-04-06.

Anthropic. (2026b). *Pricing; Prompt caching; Fast mode (research preview); Vision; Steering thinking; Migrating to Claude Opus 5.5* (developer documentation, read 28–29 September 2026). <https://platform.claude.com/docs/en/about-claude/pricing>; <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>; <https://platform.claude.com/docs/en/build-with-claude/fast-mode>; <https://platform.claude.com/docs/en/build-with-claude/vision>; <https://platform.claude.com/docs/en/build-with-claude/thinking-steering-and-cost>; <https://platform.claude.com/docs/en/models/opus-5-5/migration-guide>

Artificial Analysis. (2026). *Methodology; Performance benchmarking methodology* (read 28 September 2026). <https://artificialanalysis.ai/methodology>; <https://artificialanalysis.ai/methodology/performance-benchmarking> — third-party measurement definitions.

Azam, R., Vempaty, A., & Jagmohan., A. (2025). Reflection-Based Memory For Web navigation Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2025-06-02.

Bogdanov, I., Lung, C. H., Kunz, T., et al. (2026). FORGE: Self-Evolving Agent Memory With No Weight Updates via Population Broadcast. *Proceedings of the ACM Conference on AI and Agentic Systems (ACM CAIS '26, San Jose CA, May 26-29, 2026), research paper, pp. 292-310*. Formal: [link](#). Version read: [HTML](#), v1, 2026-05-15.

Cai, Q., Zhang, Y., Jia, X., et al. (2026). MOSS: Self-Evolution through Source-Level Rewriting in Autonomous Agent Systems. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-05-23.

Cai, T., Wang, X., Ma, T., et al. (2024). Large Language Models as Tool Makers. *International Conference on Learning Representations 2024 (ICLR 2024), Conference (main)*. Formal: [link](#). Version read: [HTML](#), v2, 2024-03-11.

Chae, H., Kim, N., Ong, K., et al. (2025). Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation. *International Conference on Learning Representations 2025 (ICLR 2025), Conference*. Formal: [link](#). Version read: [HTML](#), v2, 2025-03-29.

Chen, H. M., Guo, J., Luk, W., & Fan, H. (2026). AOSpec: Action and observation co-speculation for low-latency agent serving. [arXiv:2608.00881](https://arxiv.org/abs/2608.00881). Imperial College London.

Chen, M., Wang, J., Liu, Z., et al. (2026). From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-07-02.

Chen, Y., Arkin, J., Hao, Y., et al. (2024). PRompt Optimization in Multi-Step Tasks (PROMST): Integrating Human Feedback and Heuristic-based Sampling. *EMNLP 2024, main conference: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 3859-3920, Miami, Florida, USA, November 2024; Anthology ID 2024.emnlp-main.226*. Formal: [link](#). Version read: [HTML](#), v4, 2024-10-03.

Cheng, C. A., Nie, A., & Swaminathan., A. (2024). Trace is the Next AutoDiff: Generative Optimization with Rich Feedback, Execution Traces, and LLMs. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Main Conference Track, poster*. Formal: [link](#). Version read: [HTML](#), v2, 2024-11-01.

Cheng, Z., Liu, Z., Shan, Y., et al. (2026). Mem^2Evolve: Towards Self-Evolving Agents via Co-Evolutionary Capability Expansion and Experience Distillation. *ACL 2026: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), main conference, pages 20784-20831, San Diego, California, United States, July 2026; Anthology ID 2026.acl-long.952*. Formal: [link](#). Version read: [HTML](#), v1, 2026-04-13.

Dai, Z., He, S., Li, H., et al. (2026). Metis: Bridging Text and Code Memory for Self-Evolving Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-06-23.

Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. [arXiv:2504.13359v2](https://arxiv.org/abs/2504.13359v2). Stanford University.

Esakkiraja, E., Akhiyarov, D., Yadav, V., et al. (2026). StarHarness: Evolving Harnesses with Stratified Search for Enterprise Environments. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-08-25.

Feng, G., Mao, H., Dutta, P., & Gonzalez, J. E. (2026). Concurrency without model changes: Future-based asynchronous function calling for LLMs (AsyncFC). [arXiv:2605.15077](https://arxiv.org/abs/2605.15077). University of California, Berkeley.

Guo, Y., Lin, J., Wang, H., et al. (2025). SE-Agent: Self-Evolution Trajectory Optimization in Multi-Step Reasoning with LLM-Based Agents. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Main Conference Track, poster (decision 'Accept (poster)'*. Formal: [link](#). Version read: [PDF](#), v6, 2025-11-03.

Hajimiri, S., Aminbeidokhti, M., Dolz, J., et al. (2026). Are Online Skill and Memory Modules Always Worth Their Tokens? A Budget-Constrained Study of Web Agents. *EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing), main conference, paper ID 4950-MAIN, poster (Session 4)*. Formal: [link](#). Version read: [HTML](#), v2, 2026-08-30.

Hao, G., Long, Y., & Zhao., Z. (2026). Self-Evolving Multi-Agent Systems via Decentralized Memory (DecentMem). *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-05-21.

References (2/6)

Hattami, A. E., Chapados, N., & Pal., C. (2026). SKILL.nb: Selective Formalization and Gated Execution for Durable Agent Workflows. *FAGEN@ICML 2026 Poster (ICML 2026 workshop, venueid ICML.cc/2026/Workshop/FAGEN; not the ICML main conference)*. Formal: [link](#). Version read: [HTML](#), v1, 2026-06-06.

Hu, S., Lu, C., & Clune., J. (2025). Automated Design of Agentic Systems. *International Conference on Learning Representations 2025 (ICLR 2025), Conference track (ICLR proceedings page)*. Formal: [link](#). Version read: [HTML](#), v2, 2025-03-02.

Hua, W., Wan, M., Vadrevu, S., Nadel, R., Zhang, Y., & Wang, C. (2025). Interactive speculative planning: Enhance agent efficiency through co-design of system and user interface (ISP). *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.00079. Rutgers University; Microsoft; Google DeepMind.

Huang, Z., Wang, X., Wang, A., et al. (2026). Better, Faster, Stronger: Programmatic Skill Learning Best Reduces Agent Cost. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-08-11.

Iacob, A., Jovanović, A., Shen, W. F., et al. (2026). The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators. *arXiv preprint*. Formal: [link](#). Version read: [PDF](#), v2, 2026-06-29.

Jiang, W., Zhuang, Y., Song, C., et al. (2025). AppAgentX: Evolving GUI Agents as Proficient Smartphone Users. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v3, 2025-04-15.

Jin, R., Zhang, Z., Wang, M., et al. (2025). STELLA: Self-Evolving LLM Agent for Biomedical Research. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2025-07-01.

Jin, Y., Zhang, S., Wang, H., et al. (2026). EXG: Self-Evolving Agents with Experience Graphs. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-05-18.

Kapoor, S., Stroebel, B., Siegel, Z. S., et al. (2025). AI Agents That Matter. *Transactions on Machine Learning Research*. Formal link: not recorded. Version read: [HTML](#), v1, 2024-07-01.

Karten, S., Zhang, J., Jr, T. U., et al. (2026). Continual Harness: Online Adaptation for Self-Improving Foundation Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-05-11.

Khattab, O., Singhvi, A., Maheshwari, P., et al. (2024). DSPy: Compiling Declarative Language Model Calls into State-of-the-Art Pipelines. *International Conference on Learning Representations 2024 (ICLR 2024), main conference, spotlight poster (decision 'Accept (spotlight)')*. Formal: [link](#). Version read: [HTML](#), v1, 2023-10-05.

Kim, S., Moon, S., Tabrizi, R., Lee, N., Mahoney, M. W., Keutzer, K., & Gholami, A. (2024). An LLM compiler for parallel function calling. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, PMLR 235, 24370–24391. arXiv:2312.04511. University of California, Berkeley; International Computer Science Institute; Lawrence Berkeley National Laboratory.

Lee, S., Choi, J., Lee, J., et al. (2024). MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom 2024), research paper, pp. 1119-1133*. Formal: [link](#). Version read: [HTML](#), v3, 2024-10-16.

Li, L., Li, J., Zhao, J., et al. (2026). Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-09-24.

Li, Y., Ye, Q., Choubey, P. K., Zhang, J., & Wu, C.-S. (2026). Speculate with memory: Lossless acceleration for LLM agents. arXiv:2607.12236v1. Salesforce Research.

Li, Y., Zhang, Y., Zhang, X., et al. (2026). CODESKILL: Learning Self-Evolving Skills for Coding Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-09-25.

Liang, J., Han, J., Li, W., et al. (2026). GenericAgent: A Token-Efficient Self-Evolving LLM Agent via Contextual Information Density Maximization (V1.0). *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-04-18.

Liao, J., Shi, H., Zhou, R., et al. (2026). MemQ: Integrating Q-Learning into Self-Evolving Memory Agents over Provenance DAGs. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v3, 2026-05-14.

Lindenbauer, T., Groh, G., & Schütze., H. (2025). From Knowledge to Noise: CTIM-Rover and the Pitfalls of Episodic Memory in Software Engineering Agents. *Proceedings of the 1st Workshop for Research on Agent Language Models (REALM 2025), Vienna, July 2025, pages 411-427; Anthology ID 2025.realm-1.30*. Formal: [link](#). Version read: [HTML](#), v1, 2025-05-29.

Liu, G., Geng, S., Li, S., et al. (2025). WebCoach: Self-Evolving Web Agents with Cross-Session Memory Guidance. *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents), workshop paper in Poster and Discussion Session 1*. Formal: [link](#). Version read: [HTML](#), v2, 2026-07-22.

Liu, H., Ye, T., Gao, S., et al. (2026). SoL-Pi: Recursively Scaling Auto-Research Loops for Efficient Agent Harness. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-09-17.

Liu, J., Ye, X., Xia, P., et al. (2026). EvolveMem: Self-Evolving Memory Architecture via AutoResearch for LLM Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-05-13.

References (3/6)

Liu, L., Tang, P., Singh, K. Y., et al. (2026). ESPO: Error-Structured Prompt Optimization via Diagnose, Diversify, and Stabilize. *EMNLP 2026, Findings track (paper ID 3300-FIND; in-person poster*. Formal: [link](#). Version read: [HTML](#), v1, 2026-09-03.

Liu, Z., Shi, Z., Sang, Y., et al. (2026). Adaptive Auto-Harness: Sustained Self-Improvement for Agentic System Deployment on Open-Ended Task Streams. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-06-03.

Lou, Z., Chen, H., Li, Y., et al. (2026). Learning to Learn-at-Test-Time: Language Agents with Learnable Adaptation Policies. *ICML 2026 Workshop AIWILD, under the v1 title 'Learning to Learn-at-Test-Time: Language Agents with Learnable Adaptation Policies'*. Formal: [link](#). Version read: [HTML](#), v4, 2026-09-28.

Lu, J., Zhang, Z., Yang, F., et al. (2025). AXIS: Efficient human-agent-computer interaction with API-first LLM-based agents. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Formal: [link](#). Version read: [PDF](#), [PDF](#); version not recorded.

Lyu, B., Cong, X., Yu, H., et al. (2025). Enhancing Open-Domain Task-Solving Capability of LLMs via Autonomous Tool Integration from GitHub. *ACL 2025 (63rd Annual Meeting of the Association for Computational Linguistics), main conference, Volume 1: Long Papers, pages 17257-17277, Vienna, July 2025; Anthology ID 2025.acl-long.845*. Formal: [link](#). Version read: [HTML](#), v3, 2025-06-10.

Ma, Z., Zhao, Z., Hua, C., et al. (2026). JudgeFlow: Agentic Workflow Optimization via Block Judge. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-02-02.

Madaan, A., Tandon, N., Gupta, P., et al. (2023). Self-Refine: Iterative Refinement with Self-Feedback. *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Main Conference Track*. Formal: [link](#). Version read: [HTML](#), v2, 2023-05-25.

Nguyen, M., Nguyen, Q., & Vuong., P. (2026). Recursive Self-Evolving Agents via Held-Out Selection. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-06-17.

Opsahl-Ong, K., Ryan, M. J., Purtell, J., et al. (2024). Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs. *EMNLP 2024: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (main conference), pages 9340-9366, Miami, Florida, USA, November 2024; Anthology ID 2024.emnlp-main.525*. Formal: [link](#). Version read: [HTML](#), v2, 2024-10-06.

Ou, Y., Zhou, W., Ding, S., et al. (2025). Symbolic learning enables self-evolving agents. *AI Open, Volume 6, pp. 314-322 (2025), journal article (Elsevier B.V. on behalf of KeAi; open access, CC BY 4.0)*. Formal: [link](#). Version read: [HTML](#), v1, 2024-06-26.

Ouyang, S., Yan, J., Hsu, I. H., et al. (2026). ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. *ICLR 2026, main conference, poster*. Formal: [link](#). Version read: [HTML](#), v2, 2026-03-16.

Park, J., Cho, J., & Ahn., S. (2025). MrSteve: Instruction-Following Agents in Minecraft with What-Where-When Memory. *International Conference on Learning Representations 2025 (ICLR 2025), Conference (main), poster*. Formal: [link](#). Version read: [HTML](#), v5, 2025-04-11.

Prabhu, V., Dai, Y., Fernandez, M., et al. (2026). WALT: Web Agents that Learn Tools. *International Conference on Learning Representations 2026 (ICLR 2026), Conference track (proceedings pp. 55589-55607)*. Formal: [link](#). Version read: [PDF](#), v1, 2025-10-01.

Pryzant, R., Iter, D., Li, J., et al. (2023). Automatic Prompt Optimization with "Gradient Descent" and Beam Search. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023), main conference, pages 7957-7968; Anthology ID 2023.emnlp-main.494*. Formal: [link](#). Version read: [HTML](#), v2, 2023-10-19.

Qian, H., & Liu., Z. (2025). MetaAgent: Toward Self-Evolving Agent via Tool Meta-Learning. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2025-09-01.

Qiu, J., Juan, X., Wang, Y., et al. (2025). AgentDistill: Training-Free Agent Distillation with Generalizable MCP Boxes. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2025-06-17.

Qiu, J., Qi, X., Wang, H., et al. (2025). Alita-G: Self-Evolving Generative Agent for Agent Generation. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2025-10-27.

Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025), main conference (proceedings 'Conference' track)*. Formal: [link](#). Version read: [HTML](#), v2, 2025-02-26.

Robeyns, M., Szummer, M., & Aitchison., L. (2025). A Self-Improving Coding Agent. *ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision'*. Formal: [link](#). Version read: [HTML](#), v2, 2025-05-16.

Shao, S., Zhang, K., Li, Q., et al. (2026). Harness-R1: Learning to Edit Executable Runtime Harnesses from Agent Failure Trajectories. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-08-03.

Shinn, N., Cassano, F., Gopinath, A., et al. (2023). Reflexion: Language Agents with Verbal Reinforcement Learning. *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Main Conference Track*. Formal: [link](#). Version read: [HTML](#), v4, 2023-10-10.

Sun, Z., Liu, Z., Zang, Y., et al. (2026). SEAgent: Self-Evolving Computer Use Agent with Autonomous Learning from Experience. *ICML 2026: Proceedings of the 43rd International Conference on Machine Learning, PMLR 306:116867-116893 (volume published 2026-09-29), main conference*. Formal: [link](#). Version read: [HTML](#), v2, 2025-08-12.

References (4/6)

Suresh, S., Mak, H., Bhatnagar, S., et al. (2026). Grounding Agent Memory: Environment-Probing Curation for Enterprise Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-09-10.

Suzgun, M., Yuksekgonul, M., Bianchi, F., et al. (2026). Dynamic Cheatsheet: Test-Time Learning with Adaptive Memory. *EACL 2026: Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), pages 7080-7106, Rabat, Morocco, March 2026; main conference long paper; Anthology ID 2026.eacl-long.333*. Formal: [link](#). Version read: [HTML](#), v1, 2025-04-10.

Tham, L., Garcia, N. M. G., & Hahn., J. (2026). Internal APIs Are All You Need: Shadow APIs, Shared Discovery, and the Case Against Browser-First Agent Architectures. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-04-01.

Trivedi, H., Khot, T., Hartmann, M., et al. (2024). AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents. *ACL 2024: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), main conference, pages 16022-16076, Bangkok, Thailand, August 2024; Anthology ID 2024.acl-long.850*. Formal: [link](#). Version read: [HTML](#), v1, 2024-07-26.

Wang, G., Xie, Y., Jiang, Y., et al. (2024). Voyager: An Open-Ended Embodied Agent with Large Language Models. *Transactions on Machine Learning Research (TMLR), March 2024, journal paper (ISSN 2835-8856); certifications J2C (Journal to Conference) with event certification iclr.cc/ICLR/2025/Journal_Track*. Formal: [link](#). Version read: [HTML](#), v2, 2023-10-19.

Wang, W., Alyahya, H. A., Ashley, D. R., et al. (2024). How to Correctly do Semantic Backpropagation on Language-based Agentic Systems. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2024-12-04.

Wang, W., Piękos, P., Nanbo, L., et al. (2026). Huxley-Gödel Machine: Human-Level Coding Agent Development by an Approximation of the Optimal Self-Improving Machine. *ICLR 2026 (International Conference on Learning Representations 2026), main conference (proceedings track 'Conference')*. Formal: [link](#). Version read: [HTML](#), v3, 2025-10-29.

Wang, Z. Z., Gandhi, A., Neubig, G., et al. (2025). Inducing Programmatic Skills for Agentic Tasks. *COLM 2025 (Conference on Language Modeling), accepted paper, main conference (listed on the official "COLM 2025: Accepted Papers" page without a Spotlight badge)*. Formal: [link](#).

Wang, Z. Z., Mao, J., Fried, D., et al. (2025). Agent Workflow Memory. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), PMLR 267:63897-63911, 2025 (the PMLR page names no track; conference proceedings volume)*. Formal: [link](#). Version read: [HTML](#), v1, 2024-09-11.

Wen, H., Tian, S., Pavlov, B., et al. (2025). AutoDroid-V2: Boosting SLM-based GUI Agents via Code Generation. *Proceedings of the 23rd Annual International Conference on Mobile Systems, Applications and Services (ACM MobiSys '25), June 23-27, 2025, Anaheim, CA, USA; main research-paper track (listed on the official MobiSys 2025 Accepted Papers page, with a Best Artifact Award); pp. 223-235*. Formal: [link](#). Version read: [HTML](#), v3, 2025-05-06.

Weng, Z., Antoniadou, A., Nathani, D., et al. (2026). Group-Evolving Agents: Open-Ended Self-Improvement via Experience Sharing. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-02-04.

Winston, C., Wang, R. Y., Mirhoseini, A., et al. (2026). Agent JIT Compilation for Latency-Optimizing Web Agent Planning and Scheduling. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. Formal: [link](#). Version read: [HTML](#), v2, 2026-05-29.

Wong, M., Hsieh, K., Nath, S., et al. (2026). Skim: Speculative Execution for Fast and Efficient Web Agents. . Formal link: not recorded. Version read: [HTML](#), v2, 2026-05-19.

Wu, S., Zhao, S., Huang, Q., et al. (2024). AvaTaR: Optimizing LLM Agents for Tool Usage via Contrastive Reasoning. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Main Conference Track, poster (decision 'Accept (poster)')*. Formal: [link](#). Version read: [HTML](#), v3, 2024-10-31.

Wu, Z., Han, C., Ding, Z., et al. (2024). OS-Copilot: Towards Generalist Computer Agents with Self-Improvement. *LLMAgents @ ICLR 2024 Poster: ICLR 2024 Workshop on Large Language Model Agents*. Formal: [link](#). Version read: [HTML](#), v2, 2024-02-15.

Xia, C. S., Wang, Z., Yang, Y., et al. (2025). Live-SWE-agent: Can Software Engineering Agents Self-Evolve on the Fly?. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v3, 2025-11-24.

Xu, B., Xue, Z., Chen, D., Fu, C., Wu, C., Huang, C., Jiang, C., Fang, J., Deng, X., Chen, Y., Yao, Y., Wang, X., Shang, J., Yu, G., & Zhang, N. (2026). TokenPilot: Cache-efficient context management for LLM agents. *Findings of the Association for Computational Linguistics: EMNLP 2026 (to appear)*. arXiv:2606.17016v2. Zhejiang University; University of Electronic Science and Technology of China; Xidian University; HomologyAI.

Xu, H., Hu, J., Zhang, K., et al. (2025). SEDM: Scalable Self-Evolving Distributed Memory for Agents. *SEA @ NeurIPS 2025 (NeurIPS 2025 workshop)*. Formal: [link](#). Version read: [HTML](#), v3, 2025-09-26.

Xu, W., Han, J., Guo, M., et al. (2026). AEL: Evolving Agent Harness in Open-Ended Environments. *EMNLP 2026, Findings track (Findings of ACL: EMNLP 2026), paper 5020-FIND, presentation type 'Not Presenting'; not the EMNLP main conference; Findings volume not yet in the ACL Anthology*. Formal: [link](#). Version read: [HTML](#), v2, 2026-08-25.

Yadouni, S. E., & Li., G. (2026). TraceCompiler: Skill-Guided Mining and Compilation of LLM Agent Traces into Mostly Deterministic Workflows. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-08-03.

Yan, Z., Song, D., Zhang, H., et al. (2026). OpenSkill: Open-World Self-Evolution for LLM Agents. *EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing), main conference, paper ID 3199-MAIN, virtual presentation (session 'Virtual 2', 2026-10-25, 18:00-19:30 CET, presenter Zhiling Yan), per the official detailed program linked from 2026.emnlp.org/program/*. The ACL Anthology EMNLP 2026 volume is not yet published. Formal: [link](#). Version read: [HTML](#), v1, 2026-06-04.

Yang, Y., Bhatt, N. P., Wang, K., et al. (2026). VASO: Formally Verifiable Self-Evolving Skills for Physical AI Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-06-03.

References (5/6)

Yang, Y., Gong, Z., Huang, W., et al. (2026). SkillOpt: Executive Strategy for Self-Evolving Agent Skills. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-05-25.

Yang, Y., Jin, C., Zhao, J., et al. (2026). Act More, Decide Less: Skill-Guided Adaptive Action Chunking for Long-Horizon LLM Agents. *Findings of the Association for Computational Linguistics: EMNLP 2026*. Formal link: not recorded. Version read: [HTML](#), v1, 2026-09-02.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *The Eleventh International Conference on Learning Representations (ICLR 2023)*. arXiv:2210.03629. Princeton University; Google Research.

Yin, X., Wang, X., Pan, L., et al. (2025). Gödel Agent: A Self-Referential Agent Framework for Recursively Self-Improvement. *ACL 2025: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), main conference, pages 27890-27913, Vienna, Austria, July 2025; Anthology ID 2025.acl-long.1354*. Formal: [link](#). Version read: [HTML](#), v4, 2025-05-31.

Yuan, B., Su, Y., Song, R., et al. (2026). ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation. *Agent Skills '26 Workshop at ACM CAIS 2026 (ACM Conference on AI and Agentic Systems), San Jose, May 26, 2026*. Formal: [link](#). Version read: [HTML](#), v2, 2026-05-25.

Yuan, L., Chen, Y., Wang, X., et al. (2024). CRAFT: Customizing LLMs by Creating and Retrieving from Specialized Toolsets. *International Conference on Learning Representations 2024 (ICLR 2024), main conference, poster (decision 'Accept (poster)')*. Formal: [link](#). Version read: [HTML](#), v2, 2024-03-13.

Yuan, Y., Nayak, A., Kundu, S., & Talati, N. (2026). Agentic AI workload characterization. *Proceedings of the 2026 IEEE International Symposium on Workload Characterization (IISWC 2026)*, Boulder, CO. arXiv:2605.26297v2 (21 September 2026; arXiv title “Agentic AI workload characteristics”). University of Illinois Urbana-Champaign; Gimlet Labs; Intel.

Yuksekgonul, M., Bianchi, F., Boen, J., et al. (2025). Optimizing generative AI by backpropagating language model feedback. *Nature, vol. 639, issue 8055, pp. 609–616 (2025), Article; published online 19 March 2025 (received 12 June 2024, accepted 16 January 2025)*. Formal: [link](#).

Zelikman, E., Lorch, E., Mackey, L., et al. (2024). Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation. *COLM 2024 (Conference on Language Modeling, first edition), main conference*. Formal: [link](#). Version read: [PDF](#), v3, 2024-08-16.

Zhang, G., Fu, M., Wan, G., et al. (2025). G-Memory: Tracing Hierarchical Memory for Multi-Agent Systems. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Main Conference Track*. Formal: [link](#). Version read: [HTML](#), v2, 2025-06-16.

Zhang, H., Fan, S., Zou, H. P., et al. (2026). CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification. *COLM 2026 (Conference on Language Modeling), main conference paper*. Formal: [link](#). Version read: [HTML](#), v3, 2026-08-10.

Zhang, J., Hu, S., Lu, C., et al. (2026). Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents. *ICLR 2026 (International Conference on Learning Representations 2026), main conference (proceedings track 'Conference'), poster*. Formal: [link](#). Version read: [HTML](#), v3, 2026-03-12.

Zhang, J., Zhao, B., Yang, W., et al. (2026). Hyperagents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-03-19.

Zhang, Q., Hu, C., Upasani, S., et al. (2026). Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models. *ICLR 2026, main conference, poster (decision 'Accept (Poster)')*. Formal: [link](#). Version read: [PDF](#), v3, 2026-03-29.

Zhang, S., Wang, J., Zhou, R., et al. (2026). MemRL: Self-Evolving Agents via Runtime Reinforcement Learning on Episodic Memory. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-02-12.

Zhang, W., Zeng, L., Xiao, Y., et al. (2025). AgentOrchestra: Orchestrating Multi-Agent Intelligence with the Tool-Environment-Agent (TEA) Protocol. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v6, 2026-05-28.

Zhang, X., Zhang, W., Ng, S. K., et al. (2026). Self-Evolving World Models for LLM Agent Planning. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-09-01.

Zhang, Y., Wei, C., Du, P., et al. (2026). RewardHarness: Learning Human Preferences for Image Editing with Only 100 Demonstrations. *COLM 2026 (Third Conference on Language Modeling), main conference, poster (Poster Session 4, Imperial Ballroom #55)*. Formal: [link](#). Version read: [HTML](#), v1, 2026-05-09.

Zhang., D. (2026). AgentDevel: Reframing Self-Evolving LLM Agents as Release Engineering. *arXiv preprint*. Formal: [link](#).

Zhao, A., Huang, D., Xu, Q., et al. (2024). Expel: LLM Agents Are Experiential Learners. *Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38 No. 17: AAAI-24 Technical Tracks 17, AAAI Technical Track on Natural Language Processing II, pp. 19632-19642 (published 2024-03-24), DOI 10.1609/aaai.v38i17.29936*. Formal: [link](#). Version read: [HTML](#), v3, 2024-12-20.

Zhao, Y., Shanmugham, A., Roy, S., et al. (2026). EchoPath: Execution-Level Replayable Memory for GUI Agents. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2026-09-15.

Zhao, Z., Li, H., Zhang, H., et al. (2025). Curious Causality-Seeking Agents in Open-ended Worlds (arXiv listing title for 2506.23068: 'Curious Causality-Seeking Agents Learn Meta Causal World'). *NeurIPS 2025 (poster), main conference: Advances in Neural Information Processing Systems 38, Main Conference Track*. Formal: [link](#). Version read: [HTML](#), v3, 2025-10-26.

References (6/6)

Zheng, B., Fatemi, M. Y., Jin, X., et al. (2025). SkillWeaver: Web Agents can Self-Improve by Discovering and Honing Skills. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v1, 2025-04-09.

Zheng, L., Yin, L., Xie, Z., Sun, C., Huang, J., Yu, C. H., Cao, S., Kozyrakis, C., Stoica, I., Gonzalez, J. E., Barrett, C., & Sheng, Y. (2024). SGLang: Efficient execution of structured language model programs. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 62557–62583. <https://doi.org/10.52202/079017-2000>. arXiv:2312.07104. Stanford University; University of California, Berkeley; Shanghai Jiao Tong University; Texas A&M University; independent researcher.

Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2026-09-28.

Zhong, Y., Liu, S., Chen, J., Hu, J., Zhu, Y., Liu, X., Jin, X., & Zhang, H. (2024). DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving. *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 193–210. <https://www.usenix.org/conference/osdi24/presentation/zhong-yinmin>. arXiv:2401.09670. Peking University; StepFun; University of California, San Diego.

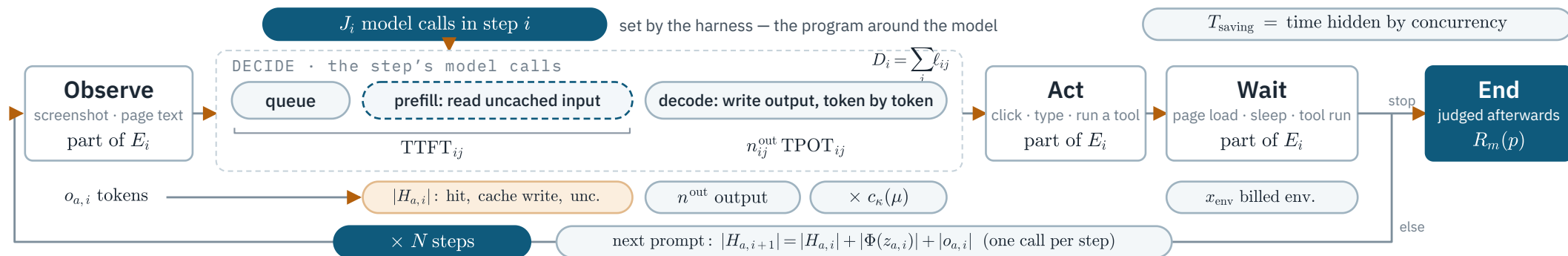
Zhou, H., Chen, Y., Guo, S., et al. (2025). Memento: Fine-tuning LLM Agents without Fine-tuning LLMs. *arXiv preprint*. Formal: [link](#). Version read: [HTML](#), v2, 2025-08-25.

Zhuge, M., Wang, W., Kirsch, L., et al. (2024). GPTSwarm: Language Agents as Optimizable Graphs. *ICML 2024: Proceedings of the 41st International Conference on Machine Learning, PMLR 235:62743-62767 (2024), main conference*. Formal: [link](#). Version read: [HTML](#), v3, 2024-08-22.

A0 · Page 1’s formulas: the original, ours, and why we changed it

OUR FORMULA	ORIGINAL FORMULA	ORIGINAL DEFINITIONS	WHAT WE CHANGED, AND WHY
$v(m, p) = \frac{C_m(p)}{R_m(p)}$ from the paper	$v(m, p) = \frac{C_m(p)}{R_m(p)} \text{ (Eq. 2)}$ (Erol et al., 2026)	m : a model; p : a problem $C_m(p)$: expected dollars of one attempt, input and output tokens times their prices (Eq. 13) $R_m(p)$: chance that one attempt is correct; $1/R_m(p)$ attempts are expected until the first correct one, attempts independent	m becomes the agent: the model plus its harness - why: learning changes the harness, not the model
$T_{\text{success}}(m, p) = \frac{\mathbb{E}[T_{\text{attempt}}]}{R_m(p)}$ self-defined, adapted	no time formula (Erol et al., 2026)	units per attempt other than dollars, such as time or latency, may matter more (App. D.1)	- the same form, in seconds - why: time and money are reported side by side - assumes attempts run one after another
$\bar{v}(m) = \frac{1}{n} \sum_{k=1}^n v(m, p_k)$ self-defined, adapted	$V_{p \sim D}(\mathcal{M}) \approx \mathbb{E}_{p \sim P}[V_p(\mathcal{M})] \text{ (Eq. 8)}$ (Erol et al., 2026)	D : a dataset of problems; P : its empirical distribution, mass $1/n$ on each problem $V_p(\mathcal{M})$: the lowest cost per success on p among the available models	- the same average over n tasks, applied to one agent’s v instead of the cheapest model’s; $\bar{T}_{\text{success}}$ and $\bar{R}$ the same way - why: we compare one agent before and after learning
$v_n(m') = \frac{C_{\text{learn}}}{n} + \bar{v}(m')$ self-defined, adapted	total cost = fixed cost + variable cost (in words) (Kapoor et al., 2025) $O(nc + C)$ against $O(nC)$ (Table 2) (Cai et al., 2024)	- fixed cost: paid once, to optimise the agent’s design (prompt, hyperparameters) - variable cost: paid on every run, set by its input and output tokens n : task instances; C : one call of the large tool-making model; c : one call of the small tool-using model	- fixed cost ÷ n : a cost per task - variable part = the average cost per success $\bar{v}(m')$, so failed attempts are paid for - fixed cost widened to exploring, proposing, checking and upkeep — why: our learning does all of these
$\min (\bar{T}_{\text{success}}(m'), v_n(m'))$ s. t. $\bar{R}(m') \geq R_0$ self-defined	$V_p(\mathcal{M}) = \min_{m \in \mathcal{M}} v(m, p) \text{ (Eq. 3)}$ (Erol et al., 2026) accuracy and cost optimised jointly (in words) (Kapoor et al., 2025)	$\mathcal{M}$: the available models $V_p(\mathcal{M})$: the lowest cost per success on p among them	- one number becomes a pair, seconds and dollars, lowered together — why: no fixed exchange rate between them - learning cost included - success floor R_0 , fixed in advance — why: an agent must not look cheaper by succeeding less
$n^* = \frac{C_{\text{learn}}}{\bar{v}(m) - \bar{v}(m')}$ self-defined	no formula (in words) (Kapoor et al., 2025)	the jointly optimised agent is cheaper in total than the default one after about 1,350 HotPotQA tasks	- written as a formula by setting $v_n(m') = \bar{v}(m)$ — why: computable from measured costs - assumes the saving per task stays the same

A0 · Every symbol in the agent loop



- i , N step (one observe–decide–act–wait round); steps in the attempt (Hua et al., 2025)
- J_i , j model calls in step i (planner, judge, retries ...); call index (self-defined)
- ℓ_{ij} latency of call j of step i (Li et al., 2026, arXiv · Salesforce Research)
- TTFT_{ij} time to the first generated token: queueing, sending, prefill of the uncached input (Zhong et al., 2024; Artificial Analysis, 2026)
- n_{ij}^{out} , TPOT_{ij} output tokens of the call, thinking included; time per output token after the first (Erol et al., 2026 (adapted); Anthropic, 2026b; Zhong et al., 2024)
- D_i , E_i model time of step i ; all its other time (observe, act, wait, harness gaps, back-off) (adapted from Hua et al., 2025; Chen et al., 2026, arXiv · Imperial)
- T_{saving} time hidden because some of it ran at the same time; 0 if strictly serial (Feng et al., 2026, arXiv · UC Berkeley; precedent Kim et al., 2024)
- $o_{a,i}$, a tokens the result or screenshot of step i adds to the prompt; agent index (Yao et al., 2023; Y. Yuan et al., 2026)
- $|H_{a,i}|$, Φ , $z_{a,i}$ length of the prompt step i 's call reads (one call per step); chat template; the call's output: thinking, message and tool-call tokens (Y. Yuan et al., 2026)
- n^{hit} , n^{w} , n^{unc} input tokens read from the cache, written to it, or uncached (Anthropic, 2026b; Xu et al., 2026; Zheng et al., 2024)
- $c_\kappa(\mu)$, κ , μ price per token of billing class κ on serving model μ (adapted from Erol et al., 2026)
- x_{env} billed environment usage (e.g. sandbox hours) (Erol et al., 2026 (adapted))
- $R_m(p)$ chance that one attempt of agent m on task p succeeds (Erol et al., 2026)

A1 · How the three columns were judged, and the totals per class

CLASS	WORKS	TIME OR MONEY OF RUNS (YES · PARTLY)	LEARNING COST (YES · PARTLY)	UNSEEN TASKS (YES · PARTLY)
Prompts and context	15	1 · 2	5 · 5	7 · 6
Experience memory	26	5 · 15	5 · 4	11 · 14
Skills and tools	34	9 · 14	6 · 11	8 · 23
Whole harness	30	8 · 4	7 · 16	5 · 23
Predicting what an action does	3	1 · 1	1 · 2	1 · 2
Descriptions, facts and preconditions	13	6 · 4	2 · 3	3 · 8
Page and place structure	7	5 · 2	1 · 3	1 · 6
Exploring and practising	9	3 · 2	1 · 3	4 · 4
Environments that check the final state (no learning)	3	1 · 2	0 · 0	3 · 0
Direction 1, each work once	93	21 · 28	19 · 33	28 · 57
Direction 2, each work once (final-state checkers excluded)	29	14 · 8	5 · 10	8 · 18

Time or money of runs — yes: measured seconds or dollars of executing tasks with what was learned; partly: tokens, steps, calls or an author’s estimate only. **Learning cost** — yes: a measured amount for learning, building or maintaining (tokens, dollars, time, compute or rollouts); partly: an estimate, iteration counts only, or a part knowingly left out. **Unseen tasks** — yes: tested on tasks that took no part in learning or selection; partly: an online stream that learns as it goes, the same workflow with new parameters, or an unstated split. These are our initial judgments from the cells of the per-work table (calc.), not re-read from the full texts; the representative works on pages 4 and 6 are to be checked one by one before the deck is final.

A02 · Prompts and context (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Reflexion (Shinn et al., 2023)	NeurIPS 2023	Text reflection added to next attempt on same task; model not updated	ALFWorld, QA, code; ALFWorld retried in same environment	ReAct GPT-3 prompt setting; ALFWorld cumulative 130 of 134 tasks; WebShop no improvement by four trials	Report 12 consecutive trials; tokens, cost, time not reported; trial count ≠ steps	○ ○ ○
Self-Refine (Madaan et al., 2023)	NeurIPS 2023	Current output and feedback history for next round	Text rewriting, code, math; per-question testing with answer changes	GPT-4 code optimization 27.3% → 36.0%; GPT-3.5 math 64.1 → 64.1; compared with same model single generation	Iteration limit 4; tokens, cost, time not reported	○ ○ ○
TextGrad (Yuksekgonul et al., 2025)	Nature 2025	Text variables; cross-question prompts or current-question solutions	Math, QA, code; prompt experiment split train/validation/test	gpt-3.5-turbo-0125 execution, gpt-4o feedback; Object Counting 77.8 → 91.9 vs DSPy 84.9; result table does not clearly mark split assignment	Prompt optimization batch size 3, 12 iterations; 1+1+1 model calls per code round; time and cost not measured	○ ● ●
MIPRO (Opsahl-Ong et al., 2024)	EMNLP 2024	Module instructions and examples	Multi-hop QA, evidence retrieval, classification; final held-out test	Llama-3-8B HotPotQA test exact match 46.4 vs unoptimized 36.1, mean of 5 runs; Iris 88.6 vs Bootstrap RS 94.1	Only trial budget evaluated; actual measured tokens, total calls, time, cost not reported	○ ● ●
ProTeGi (Pryzant et al., 2023)	EMNLP 2023	Prompt text	Classification task; train, dev, test separated, but final candidate selection by test F1 not clarified	Default gpt-3.5-turbo; Jailbreak test F1 when matching total API query budget: single selection after search 0.80, greedy 0.82, multi-candidate search 0.85; ~3 steps to peak	Author reported optimization sometimes past 1 hour, not per-setting timed table; deployment cost and time not measured	○ ● ●
Semantic Backpropagation (W. Wang et al., 2024)	arXiv	Fixed text instructions in computational graph	GSM8K, BBH, BigCodeBench; main comparison uses held-out test	gpt-4o-mini execution, gpt-4-turbo feedback update: GSM8K 93.2 vs TextGrad 78.2; BigCodeBench 27.8 vs 27.6; run count not reported	LIAR forward input tokens 273,300 vs removed neighbor info 289,850; optimization process metric, time and cost not measured	○ ● ●
Trace / OptoPrime (Cheng et al., 2024)	NeurIPS 2024	Prompts, function code or text-representable parameters	BBH held-out questions, controlled held-out initial conditions, new chess positions	BBH with reasoning template overall 78.6 vs DSPy prompt optimization 71.6; optimizer GPT-4-0125-Preview, in-program model not clearly stated	Word Sorting vs TextGrad: total experiment time 82.5 vs 211.1 min, including optimization, validation and testing; not per-question deployment latency	○ ● ●
PROMST (Chen et al., 2024)	EMNLP 2024	Action prompts; separately train Longformer scorer, not all models frozen	Web, text environments, planning; main performance independence on held-out not clarified	GPT-4 average progress score 0.69 vs PromptAgent 0.61; WebArena 0.62 vs APO 0.64; progress score ≠ full-task success rate	Time, cost, tokens or total calls not measured; some tasks with higher score need more steps, collisions also increase	○ ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A02 · Prompts and context (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Dynamic Cheatsheet (Suzgun et al., 2026)	EACL 2026	Online-maintained text, strategies and code snippets	Math, science QA; accumulated in single test stream	GPT-4o Game of 24: baseline 10%, fixed-prompt zero memory 19%, retrieved organized memory 99%; Claude 3.5 Sonnet baseline 12% → 14%	AIME 2024 Claude: 370 / 494 / 1035 / 1831 tokens; memory version higher; token types and management overhead range unclear; time and cost not measured	● ○ ●
GPTSwarm (Zhuge et al., 2024)	ICML 2024 main conference	Node prompts, inter-agent connection probability; not LLM weights	GPT-3.5/GPT-4; MMLU, Crosswords, HumanEval	Crosswords same 20 questions optimization evaluation 0.465 → 0.575; HumanEval online 0.76 → 0.88	MMLU optimization 5.32 USD, inference 1.82 USD; HumanEval with optimization total 28.46 USD vs 1.61 USD, cannot be viewed as same-scope deployment cost	● ● ●
AvaTaR (S. Wu et al., 2024)	NeurIPS 2024	Instructions and action sequences; task-direct action program reuse	Knowledge base retrieval, multi-tool QA; training then testing official and human held-out queries	claude-3-opus STaRK Amazon/MAG/Prime top-1 hit rates 49.87 / 44.36 / 18.44 vs Reflexion 42.79 / 40.71 / 14.28	Time and cost not measured; timeout-triggered rewrite is mechanism only, threshold not given	○ ○ ●
DSPy (Khattab et al., 2024)	ICLR 2024	Select prompts from successful trajectories and ensemble; fine-tuning branch not counted	GSM8K, HotPotQA; held-out test present, some report dev set only	GPT-3.5 GSM8K uncompiled vanilla test 25.2 vs CoT bootstrap+ensemble 81.6; architecture and prompt change together, not single factor	Compilation needs minutes to tens of minutes per author estimate; deployment tokens, time, cost not measured	○ ● ●
GEPA (Agrawal et al., 2026)	ICLR 2026	Module prompts only, model frozen	QA, instructions, privacy rewriting, math; train validation test separated	Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins	Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost	○ ● ●
ACE (Q. Zhang et al., 2026)	ICLR 2026	Text context entries, model frozen	AppWorld, finance; offline held-out and online task stream tested separately	DeepSeek-V3.1 AppWorld: offline labeled 59.4, unlabeled 57.2, baseline 42.4; no warm-up online 56.1; FiNER unlabeled online 67.3 < baseline 70.7	AppWorld offline adaptation 9517 s vs GEPA 53898 s; online FiNER 2.9 vs DC 17.7 dollars; adaptation cost metric; statistical scope has unexplained differences	● ● ●
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Dynamic Cheatsheet (Suzgun et al., 2026)	EACL 2026	Online-maintained text, strategies and code snippets	Math, science QA; accumulated in single test stream	GPT-4o Game of 24: baseline 10%, fixed-prompt zero memory 19%, retrieved organized memory 99%; Claude 3.5 Sonnet baseline 12% → 14%	AIME 2024 Claude: 370 / 494 / 1035 / 1831 tokens; memory version higher; token types and management overhead range unclear; time and cost not measured	● ○ ●
ACE (Q. Zhang et al., 2026)	ICLR 2026	Text context entries, model frozen	AppWorld, finance; offline held-out and online task stream tested separately	DeepSeek-V3.1 AppWorld: offline labeled 59.4, unlabeled 57.2, baseline 42.4; no warm-up online 56.1; FiNER unlabeled online 67.3 < baseline 70.7	AppWorld offline adaptation 9517 s vs GEPA 53898 s; online FiNER 2.9 vs DC 17.7 dollars; adaptation cost metric; statistical scope has unexplained differences	● ● ●
Agent Workflow Memory (AWM) (Wang et al., 2025a)	ICML 2025	Infer text workflows from trajectories; action encapsulation has variants	WebArena online; Mind2Web cross-task, site, domain	GPT-4-0613 WebArena: 35.5 vs BrowserGym 23.5 success rate; compared input contains HTML, not strict same interface; Mind2Web full-task success rate 0.7—4.8	WebArena average steps 5.9 vs author rerun with accessibility-tree-only BrowserGym 7.9; step comparison not same as 23.5 success-rate row	● ● ●
ExpeL (Zhao et al., 2024)	AAAI 2024	Suggestion list and retrieved successful trajectories	HotpotQA, ALFWorld, WebShop; train and eval tasks separated	gpt-3.5-turbo-0613 ALFWorld single-attempt success 59% vs Reflexion after third retry 54%; different test budget	Training retry limit 3 only; extraction calls, tokens, time or cost not measured	○ ○ ●
MemRL (S. Zhang et al., 2026)	arXiv	Experience repository and retrieval value based on reward	Code, OS, database, ALFWorld, HLE; repeated training vs frozen memory with held-out separated	Held-out average success rate vs MemP +2.8 pp; ALFWorld +5.8, OS +2.6; different baseline foundation models	HLE Gemini-3-pro average ~32K tokens per question across 10 epochs; whether measured per attempt unclear; no comparable MemP figures	● ● ●
MemQ (Liao et al., 2026)	arXiv	External experience value and generated relevance graph	Multimodal QA, code, tools, OS; freeze best round memory for held-out	Gemma-4-E4B-it LiveCodeBench: 51.43±2.86 vs RAG 50.48±1.65, MemRL 45.71±2.86; 3 seeds; best round selection not clarified	Time, cost, tokens or calls not measured; 'faster learning' means per-round success rate	○ ○ ●
AEL (Xu et al., 2026)		Main config learns memory retrieval selection, reflection, rules; code evolution default off	Claude Haiku 4.5 multi-task allocation, Gemini 3.1 Flash-Lite work-order stream	Multi-task allocation test Sharpe 2.13 +/- 0.47; EvoTool 1.37 +/- 1.74, but difference p=0.39; planning evolution -1.72	Author estimate full run \$2.80, prediction ~\$2, reflection evolution ~\$0.80; latency not measured	○ ● ●
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Mem²Evolve (Cheng et al., 2026)	ACL 2026	Expert agents, executable tools and experience	General assistant, math, QA, planning, embodied, shopping; main result test stream accumulation inferred as entries	GPT-5-chat ALFWorld 94.31 vs AFLOW 93.40, HotpotQA 60.80 same as AFLOW; cross-task memory analysis has separate held-out	Tool average repair rounds 1.01 → 0.48; tokens, time cost not measured; rounds not convertible to them	● ○ ●
FORGE (Bogdanov et al., 2026)	ACM CAIS 2026	Multiple individual rules or example memory, champion broadcasts entire	CybORG CAGE-2 fixed attacker; train then freeze evaluation, same scenario	Gemini best example scheme reward -24.5 vs zero-shot -189.6, closer to zero is better; other attackers not tested	Gemini rule scheme per-individual adaptation 7.3M +/- 1.6M tokens vs Reflexion 11.4M +/- 0.5M; time and cost not measured	○ ● ●
G-Memory (G. Zhang et al., 2025)	NeurIPS 2025	Multi-agent interaction, task and suggestion associative memory	QA, text embodied, planning; independent memory build vs held-out not clear	Qwen-2.5-14b + MacNet ALFWorld 58.21% → 79.10%; run count not reported	AutoGen+PDDL relative to no memory 1.4×10^6 tokens more, score 10.32 higher; management cost not separated; time and cost not measured	● ○ ●
CTIM-Rover (Lindenbauer et al., 2025)	ACL 2025 REALM workshop	General and repo-level text suggestions, can include successful trajectory examples	SWE-bench Verified, 401 train, 45 test, also Lite trajectories	GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified	Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured	● ○ ●
EvolveMem (J. Liu et al., 2026)	arXiv	Retrieval config, answer style and re-extracted memory for gaps	LoCoMo conversational QA, MemBench; main performance held-out independence not reported	GPT-4o LoCoMo F1 0.543 vs SimpleMem 0.432; evolution same question 30.5 → 54.3	Single LoCoMo sample full evolution 25-35 min; retrieval 15 ms, validation ~2-3 s additional; no end-to-end latency	● ● ●
DecentMem (Hao et al., 2026)	arXiv	Per-agent local experience pool and exploration/reuse router	Math, code, BBH, ALFWorld; held-out and retry count not reported	Qwen3-4B + AgentNet average accuracy 53.69 vs G-Memory 43.37, no memory 35.20	Qwen3-8B BBH vs G-Memory token reduction 32% / 49% / 47%, unclear if judging included; no-memory still cheaper	● ○ ●
EXG (Jin et al., 2026)	arXiv	Failure-success experience graph, connecting failures to repair cases	Code and multi-hop QA; online stream and frozen graph 7:3 held-out	Qwen3-Coder-Flash HumanEval held-out: EXG-Reflexion 0.879 vs ExpeL 0.909; online has larger gains	HumanEval baseline EXG: 1.20 vs Reflexion 1.83 model calls; LLM latency 3,259 vs 3,416 ms; total tokens 125,304 vs 117,378; cost-graph model not clear	● ○ ●
Meta-TTL (Lou et al., 2026)		Cross-round action prompts for same task and 'how to change prompt' prompt learned offline	Jericho, WebArena-Lite, tau2-bench; new instances and new domains	Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate	Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
SE-Agent (Guo et al., 2025)	NeurIPS 2025	Candidate trajectory pool for same task, rewriting, recombining and filtering	SWE-bench Verified; per-question search, not cross-question held-out memory	GPT-4o: Pass@1 40.4% vs SWE-Agent 22.4%, SWE-Search 32.6%; internal multi-candidate, not single trajectory	Default 10 candidates; max API cost graph unit unclear; actual tokens, time cost not measured	○ ○ ○
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	● ○ ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
ReAP (Azam et al., 2025)	arXiv	Text reflection retrieved by task similarity	WebArena, same-task redo vs similar new-task tested separately	AgentOccam gpt-4o-2024-08-06; 80% build, 20% test new-task setup; reflection vs no memory +11 points; old success task best only 84%	Same-task redo average steps 11.92 → 10.08; new task step reduction 29.1%; time cost not measured; build reports one trajectory per task only	● ● ●
Memento (Zhou et al., 2025)	arXiv	Case library; parameter branch additionally trains two-layer MLP retrieval scorer	GAIA and research QA; GAIA validation set builds library then test set	GPT-4.1 planning, o3 execution: GAIA validation Pass@3 87.88%, test set using only validation-accumulated library 79.40%; different protocols	Each level sample 10 questions; GAIA L1 input/output 26k/4.7k, L3 121k/9.8k tokens; not complete learning cost, no savings comparison	● ○ ●
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Grounding Agent Memory (Suresh et al., 2026)	arXiv	Only modify post-task experience library, add environment read-only probe	CLBench 40 question, unreported structure change; gpt-5.4; five paired run	Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7	Full execution \$3.38 → \$1.68; no experience-build and probe-maintain	● ○ ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	● ○ ●
Mem²Evolve (Cheng et al., 2026)	ACL 2026	Expert agents, executable tools and experience	General assistant, math, QA, planning, embodied, shopping; main result test stream accumulation inferred as entries	GPT-5-chat ALFWorld 94.31 vs AFLOW 93.40, HotpotQA 60.80 same as AFLOW; cross-task memory analysis has separate held-out	Tool average repair rounds 1.01 → 0.48; tokens, time cost not measured; rounds not convertible to them	● ○ ●
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
SkillWeaver (Zheng et al., 2025)	arXiv	Playwright function library from web exploration, includes usage notes and page prerequisites	GPT-4o WebArena; self-practice then test benchmark tasks	Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1	Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost	○ ● ●
Voyager (G. Wang et al., 2024)	TMLR 2024	Executable Minecraft skills, completed and failed task tables	GPT-4; MineDojo/Mineflayer; exploration vs new worlds no task separation	New world each unseen task 3/3 success; ReAct, Reflexion, AutoGPT 0/3 within 50 prompting iterations	'15.3× faster' is prompt-iteration ratio to unlock wood tools; seconds, tokens, dollars checked entry not reported	● ○ ●
MetaAgent (Qian et al., 2025)	arXiv	Frozen model, accumulate text experience and raw tool result knowledge base; no explicit environment-rules model	QwQ-32B GAIA/WebWalkerQA/BrowseComp; pre-fill library with same-batch task simulation	WebWalkerQA 52.1 vs WebThinker-32B-RL 46.5; GAIA 47.6 below its 48.5; no held-out split	Checked entry does not report measured cost or time	○ ○ ○
SkillOpt (Y. Yang et al., 2026)	arXiv	Deploy skill documentation; execution model and framework fixed	GPT/Qwen; QA, table, document, math, ALFWorld; independent test	GPT-5.5 direct dialogue average 58.8 → 82.3 vs no skill; best per-cell baseline average 76.9	Training tokens e.g. SpreadsheetBench 21.4M; cost per question inconsistent scope; deployment per-task usage and time not reported	○ ● ●
VASO (Y. Yang et al., 2026)	arXiv	Logic skill rules, proposition mapping, plan templates	Jackal/PX4 planning and simulation; GPT-5-nano skill generation, GPT-4o-mini planning	Automated rules satisfy 96.8, task completion 86.5; LLM-Planner 90.3/89.3; new prompt safety 92–100%, completion 80–96%	Author reported under 20 minutes per skill; skill execution time and cost not reported	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AgentOrchestra (W. Zhang et al., 2025)	arXiv	Prompts, tool/agent/memory/environment code and scheme version	GAIA, HLE, GPQA, AIME; mainly Gemini 3 Flash Preview, browse component different models	HLE 55.2 → 59.6; GAIA Validation 89.70 → 93.33; independent evolution split not stated	Author rough estimate simple task 30 seconds/5k tokens, complex 10 minutes/100k; no Vanilla/Evolved cost comparison	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
STELLA (Jin et al., 2025)	arXiv	Biomedical tool pool, reasoning template	Claude 4 Sonnet + Gemini 2.5 Pro; HLE biomedical, LAB-Bench subset	HLE 14% → 26%; LitQA ~52% → 63% with trial budget increase; clear evolution/held-out separation missing	Only trials 1x to 9x reported; time/token/cost not reported	○ ● ●
DRAFT / From Exploration to Mastery (Qu et al., 2025)	ICLR 2025	Rewrite natural-language tool documentation, do not train execution model	GPT-4o writes docs, GPT-4o/mini/Llama-3-70B use; RestBench, ToolBench	GPT-4o TMDb correct-path rate 88.00 vs ReAct 71.00; tool path includes standard subsequence, not final-state check	Checked entry does not report run and improvement call total, tokens, time, cost	○ ○ ●
CODESKILL (Y. Li et al., 2026)	arXiv	Text skill library + independently-trained SFT/RL management model	Frozen Qwen3.5-35B-A3B or GPT-5.4-mini; programming and environment repair	Qwen execution policy average 29.32 → 40.35; skill library built from test-set prerun, excluding current-task source, not strict online stream	Average reasoning steps 44.12 → 35.99; success-task scope inconsistent; manager RL ~210 hours, SFT ~20 hours	● ● ●
LATM (Cai et al., 2024)	ICLR 2024 main conference	Python tools + call examples + tool cache	GPT-4 create; GPT-3.5 Turbo/GPT-4 use; logic, bracket, sorting, schedule held-out	GPT-3.5 Turbo Dyck Language 20.4 → 92.2; GPT-4 Logical Deduction 88.8 → 86.6	Only O(nc+C) parse cost and call-budget design; time/tokens/dollars not measured	○ ● ●
Alita-G (Qiu et al., 2025b)	arXiv	Successful-trajectory tools abstracted into MCP toolbox, retrieval select	Claude-Sonnet-4 manage + GPT-4.1 browse; GAIA validation set build, retest	GAIA pass@1 75.15% → 83.03%; PathVQA/HLE build-test relationship not explained	GAIA average generated tokens 12,305 → 10,394; build cost and latency not measured	● ○ ○
OS-Copilot / FRIDAY (Z. Wu et al., 2024)	ICLR 2024 LLMagents workshop	Frozen model, update program tool library; semantic memory is design component, experiment mainly validates tools	GAIA dev set learn, private test; self-created table tasks then SheetCopilot-20 test	GPT-4 SheetCopilot-20 Pass@1 60% vs no learning 0%; GAIA Level 3 both 6.12%	GAIA Level 1 table average time 118 vs 124 s; Level 2 221 vs 239 s; no-learning cost	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
OpenAgent (Lyu et al., 2025)	ACL 2025 main conference	GitHub repo integration, Docker environment, experience summary	GPT-4, GPT-3.5; OpenAct; rerun experience experiment separate	GPT-4 51-question environment/experience reuse 47.0 → 82.3; main table first-time task 47.3 vs ReAct 24.6, average scope not fully recomputable	Time/tokens/dollars checked entry not reported	○ ○ ●
OpenSkill (Yan et al., 2026)		Per-target-task skill script + substitute test	Opus 4.6/Claude Code, GPT 5.2/Codex; SkillsBench; same-task skill build then rescore	SkillsBench no skill → OpenSkill 25.5% → 43.6%, 25.0% → 42.1%; 5 evals per skill	Opus average run 465.0 → 845.4 s; skill build ~749K tokens/39 min; per-stage includes estimates, totals not fully compatible	● ● ●
CoEvoSkills (H. Zhang et al., 2026)	COLM 2026 main conference	Skill documentation, code and co-improved substitute test	Opus 4.6/Claude Code, GPT-5.2/Codex; SkillsBench same-question improvement and test	Opus pass rate 30.6% → 71.1%, manual skill 53.5%; not held-out generalization	Average 4.1 validation cycle, 2.4 ground-truth round; seconds/tokens/dollars not measured	○ ● ●
AgentDistill (Qiu et al., 2025a)	arXiv	Extract and merge MCP toolbox from frozen teacher trajectories for frozen student	GPT-3.5, Qwen3-8B, LLaMA3.1-8B; Game of 24, medical VQA; held-out relationship not stated	GPT-3.5 Game of 24 accuracy 34.3±3.2 → 82.7±0.6; build 100 validation examples	Only tool-call proportion reported; call count, time, tokens, dollars not measured	● ○ ●
CRAFT (Yuan et al., 2024)	ICLR 2024 main conference	Abstract and retrieve Python functions from solutions	GPT-4 build tools, GPT-3.5 Turbo main execution; VQA, TabMWP, MATH algebra; train source and eval separate	GPT-3.5 TabMWP 80.6 → 88.4, but same toolbox BM25 is 89.2; GQA SAcc 35.0 → 45.4	Tool build author estimate ~\$2,500; runtime cost not measured	○ ● ●
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up	● ● ●
WALT (Prabhu et al., 2026)	ICLR 2026 main conference	New action script, parameter constraint and tool description	WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute	Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change	Steps best improve 1.4x; offline tokens, dollars, time not reported	● ● ●
SpeedRunner (Huang et al., 2026)	arXiv	Frozen-model executable skill library, add/remove/modify functions	ScienceWorld, BabyAI, Crafter; gpt-5.4-mini; 200 online task, 30 held-out, 3 seed	BabyAI from ReAct ~67% to near-perfect; Crafter all-method average achievement unlock <30%	BabyAI usage ~one eighth; output-tokens vs dollars graph scope conflict, learning total cost not reported	● ● ●
ASI (Wang et al., 2025b)	COLM 2025 main conference	Python skill add to action library	WebArena online stream, no separate train set; claude-3.5-sonnet	40.4% vs plain Claude 32.7%; average action 5.0 vs 5.6, one skill = one action	Learn tokens, dollars, time not reported; step scope not equal to raw operations	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AppAgentX (Jiang et al., 2025)	arXiv	Page/element relation memory and high-level action	AppAgent 50 task/9 app; GPT-4o; five per-question	With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7	Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear	● ○ ●
TraceCompiler (Yadouni et al., 2026)	arXiv	Data-dependency executable workflow	AppWorld one Venmo intent, three instance; author hand-translates execution	Same instance tool calls 34 → 11; hold-one-instance 15 of 21 state test pass	Tool call only; compile cost and net save not tested	● ○ ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
SKILL.nb (Hattami et al., 2026)	ICML 2026 FAGEN workshop	Text step, code, pre/post check and version library, frozen model	WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity	Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%	Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time	● ● ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non-archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●
AXIS (Lu et al., 2025)	ACL 2025 main conference	Translate UI trajectory into API-code skill	Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list	Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step	29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported	● ○ ●
Skim (Wong et al., 2026)	eligibility record: EuroSys 2027	URL template, answer structure and low-cost execution path	WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework	Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease	Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported	● ○ ●
Space (Y. Yang et al., 2026)	EMNLP 2026 Findings	Supervise policy model output multi-action block using skill boundary, deploy no skill library	ALFWorld, ScienceWorld; location entry does not list per-setting model and run count	Success improve 7.0%-31.3%, decision round best -78.9%; percentage meaning not itemized	Only decision round and training step given; no time or dollars saved	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Trace / OptoPrime (Cheng et al., 2024)	NeurIPS 2024	Prompts, function code or text-representable parameters	BBH held-out questions, controlled held-out initial conditions, new chess positions	BBH with reasoning template overall 78.6 vs DSPy prompt optimization 71.6; optimizer GPT-4-0125-Preview, in-program model not clearly stated	Word Sorting vs TextGrad: total experiment time 82.5 vs 211.1 min, including optimization, validation and testing; not per-question deployment latency	○ ● ●
GPTSwarm (Zhuge et al., 2024)	ICML 2024 main conference	Node prompts, inter-agent connection probability; not LLM weights	GPT-3.5/GPT-4; MMLU, Crosswords, HumanEval	Crosswords same 20 questions optimization evaluation 0.465 → 0.575; HumanEval online 0.76 → 0.88	MMLU optimization 5.32 USD, inference 1.82 USD; HumanEval with optimization total 28.46 USD vs 1.61 USD, cannot be viewed as same-scope deployment cost	● ● ●
AEL (Xu et al., 2026)		Main config learns memory retrieval selection, reflection, rules; code evolution default off	Claude Haiku 4.5 multi-task allocation, Gemini 3.1 Flash-Lite work-order stream	Multi-task allocation test Sharpe 2.13 +/- 0.47; EvoTool 1.37 +/- 1.74, but difference p=0.39; planning evolution -1.72	Author estimate full run \$2.80, prediction ~\$2, reflection evolution ~\$0.80; latency not measured	○ ● ●
EvolveMem (J. Liu et al., 2026)	arXiv	Retrieval config, answer style and re-extracted memory for gaps	LoCoMo conversational QA, MemBench; main performance held-out independence not reported	GPT-4o LoCoMo F1 0.543 vs SimpleMem 0.432; evolution same question 30.5 → 54.3	Single LoCoMo sample full evolution 25-35 min; retrieval 15 ms, validation ~2-3 s additional; no end-to-end latency	● ● ●
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
Symbolic Learning (Ou et al., 2025)		Prompts, tools, node flow; model weights fixed	GPT-3.5/GPT-4; QA, math, code, composition; split not explained	GPT-4 MATH 60.7 vs Agents 56.0; standard benchmark disables tools, cannot prove tool creation effective	Time and cost checked entry not reported	○ ○ ●
ADAS (Hu et al., 2025)	ICLR 2025 main conference	Fixed small framework forward() program	GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test	DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline	One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured	○ ● ●
MOSS (Cai et al., 2026b)	arXiv	OpenClaw code and tool-result mediation	DeepSeek V3.2; claweval same four-task failures and retest	Average grader score 0.2526 → 0.6100, 3 per-question; no held-out task	Only iteration-1 convergence reported; time/tokens/dollars not measured	○ ● ○

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
RewardHarness (Y. Zhang et al., 2026)	COLM 2026 main conference	Reward skill and tool-specification text library, vision model frozen	Qwen2.5-VL-7B/Gemini-2.0-Flash; image editing ranking	Qwen average 45.7 vs bare model 30.3, trained EditReward 42.0; do not treat downstream image-model training as no-training evidence	77 iteration scale only reported; tokens/seconds/dollars not measured	○ ● ●
STOP (Zelikman et al., 2024)	COLM 2024 main conference	Python improver self-improve by calling fixed model	GPT-4, GPT-3.5, Mixtral; small algorithm problems; held-out and new-task transfer	GPT-4-selected improver transfer 3SAT 21.2% → 75.1%; GPT-3.5, Mixtral average degrade	LPN ~3 thousand GPT-4 calls per round each run; deployment time dollars not measured	○ ● ●
Huxley-Gödel Machine / HGM (Wang et al., 2026)	ICLR 2026 main conference	Agent code + search-tree expand/eval schedule	SWE-bench, Polyglot; SWE GPT-5 modify, GPT-5-mini test	Same 800-eval budget SWE-Verified-60 40% → 56.7%; filter overlap SWE Lite held-out 34.8% → 40.1%	Optimization allocated CPU-hours: SWE 517 vs DGM 1231; not per-question deployment latency	○ ● ●
Gödel Agent (Yin et al., 2025)	ACL 2025 main conference	Policy code and self-modifying program	GPT-4o optimize, GPT-3.5 bounded execution; QA/math held-out test	Bounded MGSM 64.2±3.4 vs Meta Agent Search 53.4±3.5; free version calls stronger model incomparable	Full evolution author estimate ~\$15; per-question execution cost and time not measured	○ ● ●
JudgeFlow (Ma et al., 2026)	arXiv	Workflow logic blocks and configuration	gpt-4o-mini; math/code; train-test split present, candidate-scoring split not clear	MATH 58.5 vs MermaidFlow 55.4; MATH → GSM8K 92.89 vs AFlow 91.95	Single GSM8K round evaluation \$0.45, Judge \$0.01; deployment time and cost not measured	○ ● ●
Adaptive Auto-Harness (Z. Liu et al., 2026)	arXiv	Task-type branching prompts, tools, infrastructure	Sonnet4.6 execute, Opus4.6 modify; prediction market, security challenge, event forecast online stream	CTF-Dojo 37.2 → 50.2; FutureX 31.0 → 47.3, but A-Evolve 47.5	FutureX average time 244.4 → 47.6 s; PolyBench 18.2 → 42.2 s; improver tokens not saved, evolution overhead not counted	● ○ ●
Continual Harness (Karten et al., 2026)	arXiv	Online modify prompts, sub-agents, skills, memory; separate weight-training experiment	Pokémon Red/Emerald continuous no reset; Gemini series	Emerald Pro fixed 24 hours: improve from start 100% milestone, \$130 median vs minimal framework 98%, \$215; Flash-Lite improve reverse decline	Measured per-run API cost; main behavior cost is button-press, not model call; improver cost not separately reported	● ● ●
Red Queen Gödel Machine / RQGM (Iacob et al., 2026)	arXiv	Execution and evaluator code co-modify; search-score shell fixed	GPT-5.5; code, paper, proof	Polyglot held-out 71.7% vs HGM-H 69.9%, 119 vs 116 questions; each domain each baseline once	Baseline-score search tokens 1.35x-1.72x fewer; deployment per-question cost and time not measured	○ ● ●
Group-Evolving Agents / GEA (Weng et al., 2026)	arXiv	Multiple parent agents share trajectories then modify framework code	SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection	Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence	Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Hyperagents / DGM-H (Zhang et al., 2026b)	arXiv	Executor and improver shared Python program	Code, paper review, math grading; robot reward design has separate PPO training	Transfer improver to math grading imp@50=0.630, initial 0.0; 200 iterations 0.640 vs 0.610 not significant	Self-modify ~33M tokens, paper review + robot eval 0.506M/iteration; total and breakdown not fully compatible	○ ● ●
SICA (Robeyns et al., 2025)	ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop	Self-complete Python code, tools and oversight mechanism	Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task	SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out	Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000	● ● ○
Harness-R1 (Shao et al., 2026)	arXiv	Frozen target-outside execution hooks; separate engineering model train SFT+GRPO	Qwen3.5-9B; WebShop, ALFWorld, DBBench	Main table same-batch 44.3% → 53.6%; also generate from 10 failures, 1,270 held-out +8.9 +/- 1.5 pp	GPU config and training data number, actual time/tokens/dollars not measured	○ ● ●
AgentDevel (Zhang, 2026)	arXiv	Prompts, code, tool wrappers one release candidate	Sonnet4.5/Claude Code; SWE, WebArena, StableToolBench	SWE Lite 11.0% → 22.0%, Verified 15.0% → 30.0%; protocol keeps test, table naming not clear	StableToolBench 11 rounds only; time/tokens/dollars not measured	○ ● ●
Darwin Gödel Machine / DGM (Zhang et al., 2026a)	ICLR 2026 main conference	Frozen model outside Python tools and workflow; search process fixed	Sonnet3.5(New) modify, SWE same model, Polyglot o3-mini execute	SWE subset 20.0% → 50.0%; Polyglot full 14.2% → 30.7%, full includes search items	80 iterations, single SWE run ~2 weeks; post-improvement per-question cost/latency not measured	○ ● ●
Live-SWE-agent (Xia et al., 2025)	arXiv	Per-question script create; framework unchanged, discard tool after completion	SWE-bench Verified/Pro; multiple backbone	Gemini3Pro success 74.2% → 77.4%; 50-question GPT-5-Nano 44.0% → 14.0%	Same Gemini average per-question cost \$0.46 → \$0.48; no offline evolution does not mean total cost down	● ● ●
RSEA (Nguyen et al., 2026a)	arXiv	Policy, skill rules, step manual text; ReAct loop unchanged	Qwen2.5-7B ALFWorld; Qwen3-30B WebShop; independent test	ALFWorld 69.3% vs ReAct 64.6%, but below Dynamic Cheatsheet 70.7 and difference not significant	Tau-bench 190 meta+selection calls total; call scope not clear; time/tokens not measured	○ ● ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
StarHarness (Esakkiraja et al., 2026)	arXiv	Stirrup prompts, tools, skills, context, validation and end logic	ITBench / EnterpriseOps-Gym / AutomationBench, GPT-5.4	Held-out increment +31.7 / +15.1 / +29.3 pp	All-benchmark per-task inference cost estimate -17% / -53% / -29%; offline optimize cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Growing Harness (L. Li et al., 2026)	arXiv	Frozen model and tools, only modify shared executable code	BrowseComp-Plus, WebArena-Verified × gpt-oss-120b/20b, Qwen3.5-4B; 50 final task per setting, 3 runs	Five of six groups highest success, one group -0.7 pp; deployment model calls -76.0-91.8%	Deployment inference cost -74.4-98.6%, not evaluator/offline improver; no break-even	● ○ ●
SoL-Pi (H. Liu et al., 2026)	arXiv	Model fixed, modify run-framework context/tool mechanisms	EdgeBench, GPT-5.6 Sol; held-out count statement inconsistent	Full config score 44.8 → 42.0, token stream -49.0%; Performance config 44.8 → 47.2, -6.1%	Measure input/cache/output token stream; not total-search tokens, dollars or time	● ○ ●
Agent JIT Compilation (Winston et al., 2026)	ICML 2026	Reuse code plan on tools, execution dispatch	Five app, GPT-4.1/Gemini-2.5-Flash/Gemini-2.5-Pro average	Planner average 11.7 s vs Browser-Use 122.1 s; 10.4x speedup	End-to-end time includes planning; plan score only dimensionless agent rank; offline total cost location entry not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A06 · Predicting what an action does

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
WMA (Chae et al., 2025)	ICLR 2025	Train page-change prediction and scoring model; task execution model frozen	GPT-4o WebArena same-environment synthetic instruction learn; also Mind2Web train/test	WebArena 16.6% vs no-action-selection 12.8%, still below tree-search 19.2%; Mind2Web cross-task AWM+WMA 25.4% vs AWM w/o EF 15.3%	Per-instruction 140.3 s vs search 748.3 s; table API \$0.4 vs \$2.7; learn-collection cost not reported; training ~3 GPU hours total scope unclear	●●●
Curious Causality-Seeking Agents / MCG (Zhao et al., 2025)	NeurIPS 2025	Train context encoder, causal graph and change-prediction network; no language model	Chemical unseen state, Magnetic manipulator arm; CEM planning	Fork n=2 predict 63.18 ± 13.94 vs FCDL 57.82 ± 9.90; full model not every ablation winner	Only training rounds and param size tested; no time/GPU hours/intervention cost	○●●
WorldEvolver (X. Zhang et al., 2026)	arXiv	All models frozen, only modify environment model context: real change record, semantic rule, predict filter	Word2World predict, AgentBoard plan; memory cross-attempt and cross-task persist, Gemma-4-26B-A4B	ALFWorld predict exact-match 52.88 vs 3.60; plan best-of-5 26.12% vs 23.88%, first-try 19.4 below 22.4	Predict eval average 1.48 s/change vs zero-shot 1.05 s; total 5.24 vs 3.72 GPU hours; not plan-task cost	●●●
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	●○●
Meta-TTL (Lou et al., 2026)		Cross-round action prompts for same task and 'how to change prompt' prompt learned offline	Jericho, WebArena-Lite, tau2-bench; new instances and new domains	Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate	Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark	●○●
AppWorld (Trivedi et al., 2024)	ACL 2024 main conference	Build resettable apps and tasks, state checker; no agent improvement	API programming operation simulate app; Test-N and Test-C include unseen app	GPT4O ReAct TGC 48.8% / 30.2%; fixed agent benchmark score, not learning gain	Current-price estimate ReAct Test-N \$0.7/task; judge average under 0.6 seconds, only checker	●—●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A07 · Descriptions, facts and preconditions (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	
CTIM-Rover (Lindenbauer et al., 2025)	ACL 2025 REALM workshop	General and repo-level text suggestions, can include successful trajectory examples	SWE-bench Verified, 401 train, 45 test, also Lite trajectories	GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified	Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured	
Meta-TTL (Lou et al., 2026)		Cross-round action prompts for same task and 'how to change prompt' prompt learned offline	Jericho, WebArena-Lite, tau2-bench; new instances and new domains	Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate	Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark	
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	
MetaAgent (Qian et al., 2025)	arXiv	Frozen model, accumulate text experience and raw tool result knowledge base; no explicit environment-rules model	QwQ-32B GAIA/WebWalkerQA/BrowseComp; pre-fill library with same-batch task simulation	WebWalkerQA 52.1 vs WebThinker-32B-RL 46.5; GAIA 47.6 below its 48.5; no held-out split	Checked entry does not report measured cost or time	
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	
DRAFT / From Exploration to Mastery (Qu et al., 2025)	ICLR 2025	Rewrite natural-language tool documentation, do not train execution model	GPT-4o writes docs, GPT-4o/mini/Llama-3-70B use; RestBench, ToolBench	GPT-4o TMDB correct-path rate 88.00 vs ReAct 71.00; tool path includes standard subsequence, not final-state check	Checked entry does not report run and improvement call total, tokens, time, cost	

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

A07 · Descriptions, facts and preconditions (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AutoDroid-V2 (Wen et al., 2025)	ACM MobiSys 2025	App documentation, synthetic script and local model weight	DroidTask 158 task/13 app; Llama3.1 8B fine-tune; per-author explore and synthesis not look test set	54.4% vs same fine-tune step-by-step AutoDroid 43.9%	Phone model inference 46.3 s vs 669.2 s; does not include UI execution; offline prep and fine-tune separate	● ● ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
Unbrowse / Shadow APIs (Tham et al., 2026)	arXiv	Shared interface graph, parameter structure, response format, auth description	94 domain read, single machine, hot cache third pass, include only aligned-output fields	Average request time 950 ms vs Playwright 3,404 ms	Measure end-to-end request time; 20-new-domain cold discover median 8,200 ms, publish success 18/20; no task cost	● ● ●
SKILL.nb (Hattami et al., 2026)	ICML 2026 FAGEN workshop	Text step, code, pre/post check and version library, frozen model	WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity	Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%	Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time	● ● ●
Grounding Agent Memory (Suresh et al., 2026)	arXiv	Only modify post-task experience library, add environment read-only probe	CLBench 40 question, unreported structure change; gpt-5.4; five paired run	Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7	Full execution \$3.38 → \$1.68; no experience-build and probe-maintain	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A08 · Page and place structure

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
CTIM-Rover (Lindenbauer et al., 2025)	ACL 2025 REALM workshop	General and repo-level text suggestions, can include successful trajectory examples	SWE-bench Verified, 401 train, 45 test, also Lite trajectories	GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified	Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured	● ○ ●
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●
Continual Harness (Karten et al., 2026)	arXiv	Online modify prompts, sub-agents, skills, memory; separate weight-training experiment	Pokémon Red/Emerald continuous no reset; Gemini series	Emerald Pro fixed 24 hours: improve from start 100% milestone, \$130 median vs minimal framework 98%, \$215; Flash-Lite improve reverse decline	Measured per-run API cost; main behavior cost is button-press, not model call; improver cost not separately reported	● ● ●
MrSteve (Park et al., 2025)	ICLR 2025	Save location, event, time and access graph; additionally train VPT-Nav, not all models frozen	Minecraft pixel control, sparse resource and long-horizon navigation; Steve-1 execute	Same sequence A-B-A task repeat A step count reduce; long-horizon ~70 tasks vs baselines <10, exact values in graph only	Memory query 11.24 ms vs FIFO 438.46 ms, not task time; navigator ~23 single-GPU training hours	● ● ●
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up	● ● ●
AppAgentX (Jiang et al., 2025)	arXiv	Page/element relation memory and high-level action	AppAgent 50 task/9 app; GPT-4o; five per-question	With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7	Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear	● ○ ●
Skim (Wong et al., 2026)	eligibility record: EuroSys 2027	URL template, answer structure and low-cost execution path	WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework	Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease	Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A09 · Exploring and practising (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
SkillWeaver (Zheng et al., 2025)	arXiv	Playwright function library from web exploration, includes usage notes and page prerequisites	GPT-4o WebArena; self-practice then test benchmark tasks	Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1	Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost	○ ● ●
Voyager (G. Wang et al., 2024)	TMLR 2024	Executable Minecraft skills, completed and failed task tables	GPT-4; MineDojo/Mineflayer; exploration vs new worlds no task separation	New world each unseen task 3/3 success; ReAct, Reflexion, AutoGPT 0/3 within 50 prompting iterations	'15.3× faster' is prompt-iteration ratio to unlock wood tools; seconds, tokens, dollars checked entry not reported	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
OS-Copilot / FRIDAY (Z. Wu et al., 2024)	ICLR 2024 LLMAgents workshop	Frozen model, update program tool library; semantic memory is design component, experiment mainly validates tools	GAIA dev set learn, private test; self-created table tasks then SheetCopilot-20 test	GPT-4 SheetCopilot-20 Pass@1 60% vs no learning 0%; GAIA Level 3 both 6.12%	GAIA Level 1 table average time 118 vs 124 s; Level 2 221 vs 239 s; no-learning cost	● ○ ●
Planning to Explore / CovQValue (Amayuelas et al., 2026)	arXiv	Frozen model, store code-branch coverage graph and test history, change test-candidate selection	RepoExploreBench/TestGenEval Lite; each program independent empty library, no cross-task held-out	Gemini 3 Flash TestGenEval Lite N=24 execution average branch 108.9 vs Greedy 61.5; coverage increase but more test fail	Execution count equal, model calls not equal; no total time/tokens/cost	○ ○ ○
SEAgent (Sun et al., 2026)	ICML 2026 main conference	Update software guide and curriculum, co-train actor and interface judge, beyond directional bound	UI-TARS-7B OSWorld app subset; checked entry does not report held-out detail	Success 11.3% → 34.5%, latter first specialize then distill then generalist train; direct all-app train 30.6%, mean of three	1k train iterations, 8 A100 80GB; no GPU hours, total time, tokens	○ ● ●
WALT (Prabhu et al., 2026)	ICLR 2026 main conference	New action script, parameter constraint and tool description	WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute	Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change	Steps best improve 1.4x; offline tokens, dollars, time not reported	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A09 · Exploring and practising (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AXIS (Lu et al., 2025)	ACL 2025 main conference	Translate UI trajectory into API-code skill	Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list	Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step	29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported	● ○ ◐

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up	● ● ●
WALT (Prabhu et al., 2026)	ICLR 2026 main conference	New action script, parameter constraint and tool description	WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute	Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change	Steps best improve 1.4x; offline tokens, dollars, time not reported	● ● ●
SpeedRunner (Huang et al., 2026)	arXiv	Frozen-model executable skill library, add/remove/modify functions	ScienceWorld, BabyAI, Crafter; gpt-5.4-mini; 200 online task, 30 held-out, 3 seed	BabyAI from ReAct ~67% to near-perfect; Crafter all-method average achievement unlock <30%	BabyAI usage ~one eighth; output-tokens vs dollars graph scope conflict, learning total cost not reported	● ● ●
ASI (Wang et al., 2025b)	COLM 2025 main conference	Python skill add to action library	WebArena online stream, no separate train set; claude-3.5-sonnet	40.4% vs plain Claude 32.7%; average action 5.0 vs 5.6, one skill = one action	Learn tokens, dollars, time not reported; step scope not equal to raw operations	● ○ ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
StarHarness (Esakkiraja et al., 2026)	arXiv	Stirrup prompts, tools, skills, context, validation and end logic	ITBench / EnterpriseOps-Gym / AutomationBench, GPT-5.4	Held-out increment +31.7 / +15.1 / +29.3 pp	All-benchmark per-task inference cost estimate -17% / -53% / -29%; offline optimize cost not reported	● ○ ●
AutoDroid-V2 (Wen et al., 2025)	ACM MobiSys 2025	App documentation, synthetic script and local model weight	DroidTask 158 task/13 app; Llama3.1 8B fine-tune; per-author explore and synthesis not look test set	54.4% vs same fine-tune step-by-step AutoDroid 43.9%	Phone model inference 46.3 s vs 669.2 s; does not include UI execution; offline prep and fine-tune separate	● ● ●
AppAgentX (Jiang et al., 2025)	arXiv	Page/element relation memory and high-level action	AppAgent 50 task/9 app; GPT-4o; five per-question	With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7	Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Growing Harness (L. Li et al., 2026)	arXiv	Frozen model and tools, only modify shared executable code	BrowseComp-Plus, WebArena-Verified × gpt-oss-120b/20b, Qwen3.5-4B; 50 final task per setting, 3 runs	Five of six groups highest success, one group -0.7 pp; deployment model calls -76.0-91.8%	Deployment inference cost -74.4-98.6%, not evaluator/offline improver; no break-even	● ○ ●
TraceCompiler (Yadouni et al., 2026)	arXiv	Data-dependency executable workflow	AppWorld one Venmo intent, three instance; author hand-translates execution	Same instance tool calls 34 → 11; hold-one-instance 15 of 21 state test pass	Tool call only; compile cost and net save not tested	● ○ ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
Unbrowse / Shadow APIs (Tham et al., 2026)	arXiv	Shared interface graph, parameter structure, response format, auth description	94 domain read, single machine, hot cache third pass, include only aligned-output fields	Average request time 950 ms vs Playwright 3,404 ms	Measure end-to-end request time; 20-new-domain cold discover median 8,200 ms, publish success 18/20; no task cost	● ● ●
SKILL.nb (Hattami et al., 2026)	ICML 2026 FAGEN workshop	Text step, code, pre/post check and version library, frozen model	WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity	Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%	Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time	● ● ●
Grounding Agent Memory (Suresh et al., 2026)	arXiv	Only modify post-task experience library, add environment read-only probe	CLBench 40 question, unreported structure change; gpt-5.4; five paired run	Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7	Full execution \$3.38 → \$1.68; no experience-build and probe-maintain	● ○ ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non-archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	○ ● ●
Sol-Pi (H. Liu et al., 2026)	arXiv	Model fixed, modify run-framework context/tool mechanisms	EdgeBench, GPT-5.6 Sol; held-out count statement inconsistent	Full config score 44.8 → 42.0, token stream -49.0%; Performance config 44.8 → 47.2, -6.1%	Measure input/cache/output token stream; not total-search tokens, dollars or time	● ○ ●
AXIS (Lu et al., 2025)	ACL 2025 main conference	Translate UI trajectory into API-code skill	Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list	Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step	29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported	● ○ ●
Agent JIT Compilation (Winston et al., 2026)	ICML 2026	Reuse code plan on tools, execution dispatch	Five app, GPT-4.1/Gemini-2.5-Flash/Gemini-2.5-Pro average	Planner average 11.7 s vs Browser-Use 122.1 s; 10.4x speedup	End-to-end time includes planning; plan score only dimensionless agent rank; offline total cost location entry not reported	● ○ ●
Skim (Wong et al., 2026)	eligibility record: EuroSys 2027	URL template, answer structure and low-cost execution path	WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework	Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease	Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported	● ○ ●
Agent Workflow Memory (AWM) (Wang et al., 2025a)	ICML 2025	Infer text workflows from trajectories; action encapsulation has variants	WebArena online; Mind2Web cross-task, site, domain	GPT-4-0613 WebArena: 35.5 vs BrowserGym 23.5 success rate; compared input contains HTML, not strict same interface; Mind2Web full-task success rate 0.7—4.8	WebArena average steps 5.9 vs author rerun with accessibility-tree-only BrowserGym 7.9; step comparison not same as 23.5 success-rate row	● ● ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SkillWeaver (Zheng et al., 2025)	arXiv	Playwright function library from web exploration, includes usage notes and page prerequisites	GPT-4o WebArena; self-practice then test benchmark tasks	Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1	Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost	○ ● ●
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
Harness-R1 (Shao et al., 2026)	arXiv	Frozen target-outside execution hooks; separate engineering model train SFT+GRPO	Qwen3.5-9B; WebShop, ALFWorld, DBBench	Main table same-batch 44.3% → 53.6%; also generate from 10 failures, 1,270 held-out +8.9 +/- 1.5 pp	GPU config and training data number, actual time/tokens/dollars not measured	○ ● ●
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

D · Evidence behind the five difficulties (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	● ○ ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5- medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39- 101 tasks and missing warm-up	● ● ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non- archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

D · Evidence behind the five difficulties (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	○ ● ●
GEPA (Agrawal et al., 2026)	ICLR 2026	Module prompts only, model frozen	QA, instructions, privacy rewriting, math; train validation test separated	Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins	Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost	○ ● ●
SICA (Robeyns et al., 2025)	ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop	Self-complete Python code, tools and oversight mechanism	Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task	SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out	Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000	● ● ○
ADAS (Hu et al., 2025)	ICLR 2025 main conference	Fixed small framework forward() program	GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test	DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline	One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured	○ ● ●
Group-Evolving Agents / GEA (Weng et al., 2026)	arXiv	Multiple parent agents share trajectories then modify framework code	SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection	Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence	Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).